

Again, finiteness and split aspect in Chinese languages^{*}

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Abstract

This paper motivates a hierarchical, size-based view of finiteness in languages without inflectional morphology, such as Chinese. Drawing on an array of cross-clausal dependencies of ‘again’-type and aspectual elements in Mandarin and Cantonese, we argue that finiteness distinction is best understood as properties arising from clause size differences rather than a simple featural distinction. Empirically, Mandarin preverbal *you* ‘again’ and Cantonese postverbal *-faan* ‘again’ exhibit exceptional scopal behavior, which is not found with other ‘again’-type elements that are lower than outer aspect. Yet, this behavior is possible only across nonfinite clauses with a size smaller than TP (i.e., vP), resembling restructuring crosslinguistically. Adopting the split aspect approach and a hierarchy of complement clause sizes, we propose that *you* and *-faan* move to and agree with an outer AspP above vP, respectively. The locality constraint stems from intervention by an outer AspP in the complement clause, which is present in larger nonfinite clauses such as TPs in addition to full-fledged finite CPs. Furthermore, we argue that the minimal size of nonfinite clauses is vP, which contains an inner AspP and blocks restructuring-like patterns with inner aspectual elements. Our findings uncover a fine-grained cartography of ‘again’-type elements and aspectual elements in Chinese languages on the one hand and support a gradient distinction in finiteness marked by complement size over an all-or-nothing dichotomy on the other.

Keywords *again* • exceptional scopal behavior • finiteness • implicational complementation hierarchy • restructuring • split aspect approach • Chinese

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1 Introduction

Various predicates take clausal complements, yet not all clausal complements have equal status. Since the early discussion of restructuring pioneered by Rizzi (1976, 1978, 1993), Rouveret & Vergnaud (1980), and Burzio (1981) among many others, it has been observed that some clauses exhibit reduced clausehood and thus a higher level of syntactic transparency. This reduced clausehood has been correlated with a wide range of syntactic phenomena, including Case assignment, clitic climbing, long-distance agreement, long object movement, scrambling, and so on (e.g., Wurmbrand 1998, 2001, Cinque 2004, 2006). Despite the variety in syntactic phenomena, the underlying intuition remains clear: certain clausal complements are structurally less independent, allowing them to behave more like extensions of their matrix clauses than as standalone ones.

This paper takes this foundational observation as its point of departure. In languages where these phenomena are well-documented, such as Germanic and Romance languages, restructuring has been closely tied to nonfiniteness, often characterized by the absence of tense marking and correlated with binding and raising (Chomsky 1973, 1981 et seq.). While efforts have been made to account for this *finiteness distinction* by positing a dedicated feature like $[\pm\text{FINITE}]$ or $[\pm\text{TENSE}]$ (e.g., Chomsky 1981, Baker 2003), a *reduced clause size* approach crystallized as a key theoretical insight over the past few decades (e.g., Stowell 1992, Cinque 2004). This approach proposes that nonfinite clauses lack certain functional projections such as CP and/or even TP; thus, they are smaller in size, deriving their reduced clausehood and consequent syntactic transparency.

Existing accounts for finiteness raise an intriguing question for Chinese languages like Mandarin and Cantonese, which appear to lack overt markers for all aforementioned morphosyntactic features: Is there evidence for a finiteness distinction or its analog in Chinese languages? This paper aligns with the position advocated by C.-T. J. Huang (1982, 1989, 1998, 2022), Y.-H. A. Li (1985, 1990), Sybesma (2007, 2019), Lin (2011, 2015), Zhang (2016, 2019), Grano (2017), N. Huang (2018, 2024), He (2020, 2024), Liao & Wang (2021), and D. Li (2024), which argues that a syntactic finiteness distinction exists in Chinese languages (contra Y. Huang 1994, Hu, Pan & Xu 2001, L. Xu 2003, etc.). As for its exact manifestation, we follow Grano (2014), N. Huang (2018), and C.-T. J. Huang (2022) in proposing that *clause size* plays a crucial role in finiteness distinction, and (solely) relying on a dedicated binary finiteness feature is not sufficient (unlike what Lin (2011) and Zhang (2019) propose, for example). We argue that finiteness is manifested as a hierarchy of clause sizes, a byproduct of the capability of predicates to take clausal complements of varying sizes in (1). Furthermore, we identify the lower bound of nonfinite clauses as vP, usually regarded as a restructuring size (Wurmbrand 2001, Cinque 2006). It contains a low aspectual projection ($\text{AspP}_{\text{inner}}$) (MacDonald 2008, Tsai 2008, Travis 2010) and essentially lacks a high aspectual projection ($\text{AspP}_{\text{outer}}$), a specific clause size we refer to as *Aspect restructuring* since it is structurally “deficient” in aspect.¹

¹ As correctly pointed out by a reviewer, the term *restructuring* has also been used for phenomena across TP-sized (or even CP-sized) nonfinite clauses (e.g., Wurmbrand 2014a, N. Huang 2018). As a descriptive label, *restructuring* refers to clause union phenomena; when narrowly defined, it may refer to a specific analysis of VP-sized complementation (*lexical restructuring* in the sense of Wurmbrand 2001, 2004) or even a reanalysis of biclausal structures to monoclausal structures with an auxiliary taking a vP (*functional restructuring* in the sense of Wurmbrand 2001, 2004 and Cinque 2006; cf. Grano 2015, 2017 for both possibilities in Chinese). We follow the stricter definition in Wurmbrand 2001: 3 (and see also her Chapters 2 and 5), which distinguishes between restructuring VP clauses (= our vP) and nonrestructuring nonfinite TPs, where the former is more transparent to more cross-clausal dependencies than the latter. Having this distinction in mind, henceforth, we use *Aspect restructuring* and *restructuring* interchangeably throughout this paper.

(1) *Verb complement clause sizes in Chinese languages*

- a. $V [CP \dots [TP \dots [AspP_{outer} \dots [vP \dots [AspP_{inner} \dots [VP \dots]]]]]]$ *type I (finite)*
- b. $V [TP \dots [AspP_{outer} \dots [vP \dots [AspP_{inner} \dots [VP \dots]]]]]$ *type II (nonfinite without Aspect restructuring)*
- c. $V [vP \dots [AspP_{inner} \dots [VP \dots]]]$ *type III (nonfinite with Aspect restructuring)*

As demonstrated by N. Huang (2018) and C.-T. J. Huang (2022), the three types exhibit distinct locality profiles in Chinese. While certain dependencies can apply across nonfinite TPs and vPs but not finite CPs, such as passivization and internal topicalization, there is another type of dependency that can apply across restructuring vPs but not nonfinite TPs and finite CPs, which is the focus of this paper. We also address a tension raised by the semi-complementizer *shuo* ‘(lit.) say’ in nonfinite clauses (N. Huang 2018) along with its controversial categorical status (e.g., Ye 2024). We propose that the tension can be resolved by adopting Pesetsky’s (2021) and C.-T. J. Huang’s (2022) Exfoliation, which removes the CP layer in *shuo*-nonfinite clauses.

This gradient structural view is consistent with the *implicational complementation hierarchy* (ICH) proposed by Wurmbrand & Lohninger (2023) and inspired by Givón 1980, which suggests that clause types form a continuum, and their associated properties follow a unidirectional progression as in (2). Our proposal ultimately supports the claim by C.-T. J. Huang (2022) and He (2024) that the ICH is systematically manifested in Mandarin, and as a novel contribution to the best of our knowledge, we extend this claim to Cantonese.

(2) *Implicational complementation hierarchy*

MOST INDEPENDENT		LEAST INDEPENDENT
LEAST TRANSPARENT	proposition > situation > event	MOST TRANSPARENT
LEAST INTEGRATED		MOST INTEGRATED

(Wurmbrand & Lohninger 2023: 188)

Our argument relies on the exceptional scopal behavior of ‘again’-type elements, found in Mandarin with preverbal *you* ‘again’ (first reported by Liu (2021, 2024) and C.-T. J. Huang (2022); cf. T. Xu 2012, 2016) and in Cantonese with postverbal *-faan* ‘again’. This exceptional scopal behavior in Mandarin is shown in the naturally occurring example (3).² Here, *you* is positioned before the matrix predicate and separated from the embedded clause, but from the provided context, we see that *you* is modifying the embedded predicate rather than the matrix one.

² Mandarin examples are transcribed using *Hanyu Pinyin*, and Cantonese examples are transcribed using *Jyutping* (the Linguistic Society of Hong Kong Cantonese Romanization Scheme, 1993). Tones are represented only when necessary. Glossing abbreviations follow the Leipzig Glossing Rules with these additions: CONT = continuative, DISP = disposal, EXP = experiential, PC = phase complement, SFP = sentence-final particle.

(3) *Mandarin preverbal ‘again’ skipping*

(Tongxue: *Tamen rang wo you xiangxin aiqing*. [...] *Xinan Daxue dikeyuan daxuesheng Wang Jingkai, kandao liuchuan zai Renren Wang shang de hunli zhaopian hou, zhuanmen liuyan zhufu zhe dui xinren changchangjiujiu*: “Yinwei tamen, ...)

[(Student: They let me again believe in love. [...] Jingkai Wang, a student at the School of Earth Sciences of Southwest University, left a message wishing the couple a long and happy life together after seeing the wedding photos circulating on *Renren.com*: “Because of them, ...)]

“**You** rang wo [xiangxin aiqing] le!”

again let 1SG believe love PRF

“‘[They] let me again believe in love!’”

Not: “‘[They] again let me believe in love!’”

(Ifeng.com 2012)

To felicitously utter the sentence presented in (3), it is not necessary for there to have been a prior entity or event that caused the speaker to believe in love: simply the fact that the speaker believed in love before would be enough.³ In other words, the repeated event is the act of believing rather than the act of being made to believe, even though on the surface, *you* does not occupy a position in which it could modify ‘believe’.

Similar exceptional scopal behavior is observed with postverbal *-faan* ‘again’ in Cantonese but in the opposite direction: embedded ‘again’ can take exceptional scope over the matrix predicate. As shown in the naturally occurring example (4), although the verbal suffix *-faan* is attached to ‘go’ embedded under ‘want’, it expresses the reading that ‘I again want to go’. Since the preceding context makes it clear that the speaker wanted to but did not go to Australia, it is a case where *-faan* takes wide scope over the matrix ‘want’ instead of the embedded ‘go’, which is also indicated by the co-occurring adverb *jau* ‘again’ above ‘want’.⁴

(4) *Cantonese postverbal ‘again’ lowering*

(*Mei gitfan zicin jyunbun hai plan-zo jiu heoi jat zyun Ouzau~ dang matjyut leoihang >< batgwo hou gwai aa3~~~ [...] [Can]cel-zo laa3~~~ Ngo m-sedak heoi aa3 Batgwo ...*)

[(Before getting married, we originally planned to go around Australia~ as a honeymoon trip >< but it’s very expensive~~~ [...] We cancelled it~~~ I can’t bear to go But ...)]

M-zi wui-m-wui zihau mau jat jat moulala (jau) soeng [heoi-**faan**].

not-know will-not-will later some one day suddenly again want go-again

‘[I] don’t know if [I]’ll suddenly again want to go someday.’

Not: ‘[I] don’t know if [I]’ll suddenly want to again go someday.’

(ESDlife WOW 2014, with *jau* made optional)

We label the Mandarin case as ‘again’ *skipping* and the Cantonese case as ‘again’ *lowering*, and we schematize the two cases in (5) below.

(5) a. *Mandarin preverbal ‘again’ skipping*

HIGH PRONUNCIATION

...[‘again’ [V_{matrix} [V_{embedded} ...

LOW INTERPRETATION

³ It is important to note that the paraphrase of this presupposition is explicitly and literally stated in the context, as in *tamen rang wo you xiangxin aiqing* ‘they let me again believe in love’. We hence believe that it is more appropriate to interpret (3) as indicating repeated belief rather than repeated causation. Our Mandarin consultants also confirmed this judgment.

⁴ The original example has *jau* ‘again’, but the sentence is equally felicitous without it under the same context.

Table 1 Exceptional scopal behavior of ‘again’-type elements across different clause boundaries in Chinese languages

	Nonfinite vP	Nonfinite TP	Finite CP
Mandarin <i>you</i>	✓	✗	✗
Cantonese <i>-faan</i>	✓	✗	✗
Mandarin <i>zai</i>	✗	✗	✗
Cantonese <i>-gwo</i>	✗	✗	✗

b. *Cantonese postverbal ‘again’ lowering*

HIGH INTERPRETATION

$$\dots[V_{\text{matrix}} [V_{\text{embedded}}\text{-‘again’} \dots]$$

LOW PRONUNCIATION

These patterns we generalize are not a Chinese-specific phenomenon. As reported by [Cardinaletti \(2003\)](#) and [Liu \(2023\)](#), Italian *ri-* ‘again’ exhibits a similar scope “skipping” behavior like Mandarin *you*, as illustrated by the narrow scope reading in (6). Crucially, this behavior is observed only with restructuring verbs (which allow clitic climbing) ([Liu 2023](#)).

(6) *Italian ri-separation*Gianni **ri**-vorrà [chiudere la porta].

Gianni again-want.FUT.3SG close.INF the door

Wide scope: ‘Gianni will again want to close the door.’

Narrow scope: ‘Gianni will want to again close the door.’

([Liu 2023](#): 2)

The empirical focus of this paper is on Mandarin and Cantonese, where restructuring has received relatively less attention. Based on naturalistic examples on the internet, data reported in the literature, as well as those we systematically collected from 9 other native speakers of (Mainland) Mandarin (with some core data confirmed by 4 more) and 9 others of (Hong Kong) Cantonese, we discover that this exceptional scopal behavior is sensitive to two syntactic factors: (i) it is allowed with certain ‘again’-type elements — namely, Mandarin *you* and Cantonese *-faan* but not with Mandarin *zai* and Cantonese *-gwo*, where the latter two we argue to be structurally lower; (ii) it is allowed with nonfinite clause-taking predicates but not with finite clause-taking ones, as summarized in [Table 1](#). As we will demonstrate, not every nonfinite clause is transparent for this behavior. Only nonfinite clauses that are vP-sized (i.e., Aspect restructuring clauses) allow for the exceptional scopal behavior, whereas TP-sized nonfinite clauses systematically block it.

To account for the dual asymmetries, we adopt the split aspect approach ([MacDonald 2008](#), [Tsai 2008](#), [Travis 2010](#), [Lu, Lipták & Sybesma 2019](#), [Yip 2020, 2025](#), [S.-W. Tang 2022](#)) and assume that there are two positions associated with Mandarin and Cantonese ‘again’-type elements: an outer AspP (AspP_{outer}) above vP and an inner AspP (AspP_{inner}) below vP, as schematized in (7). Both preverbal *you* and postverbal *-faan* associate syntactically with AspP_{outer}, via movement for *you* and via agreement for *-faan*. *Zai* and *-gwo*, in contrast, do not have an association with AspP_{outer} but only with AspP_{inner}.

- (7) a. *Association with AspP_{outer} by movement*

$$[\text{AspP}_{\text{outer}} \text{ you AspP}_{\text{outer}} [\text{vP} \dots [\text{AspP}_{\text{inner}} \dots [\text{vP} \text{ t V} \dots]]]]$$

 b. *Association with AspP_{outer} by agreement*

$$[\text{AspP}_{\text{outer}} \text{ AspP}_{\text{outer}} [\text{vP} \dots [\text{AspP}_{\text{inner}} \dots [\text{vP} \text{ V-faan} \dots]]]]$$

We propose that it is this very association with AspP_{outer} that derives the exceptional scopal behavior. In vP-sized restructuring clauses which lack AspP_{outer}, *you* and *-faan* must be licensed by the matrix AspP_{outer}. In the case of *you*, it overtly moves to AspP_{outer} but is interpreted at LF in its base-generated position, whereas embedded *-faan* agrees with the matrix AspP_{outer} to obtain wide scope.

- (8) a. *Exceptional narrow scope by reconstruction (reconstruction site indicated by a framebox)*

$$[\text{AspP}_{\text{outer}} \text{ you AspP}_{\text{outer}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{embedded}} \text{ t V} \dots]]]]$$

 b. *Exceptional wide scope by agreement*

$$[\text{AspP}_{\text{outer}} \text{ AspP}_{\text{outer}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{embedded}} \text{ V-faan} \dots]]]]$$

In the case of TP-sized nonfinite clauses, embedded AspP_{outer} serves as an intervener (in the sense of Chomsky 2000) that blocks the association with matrix AspP_{outer}.

- (9) a. *Intervention to movement to matrix AspP_{outer} (intervener indicated by shade)*

$$*[\text{AspP}_{\text{outer}} \text{ you AspP}_{\text{outer}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{matrix}} \dots [\text{TP}_{\text{embedded}} \dots [\text{AspP}_{\text{outer}} \text{ AspP}_{\text{outer}} [\text{vP}_{\text{embedded}} \text{ t V} \dots]]]]]]$$

 b. *Intervention to agreement with matrix AspP_{outer} (intervener indicated by shade)*

$$*[\text{AspP}_{\text{outer}} \text{ AspP}_{\text{outer}} \dots [\text{vP}_{\text{matrix}} \dots [\text{vP}_{\text{matrix}} \dots [\text{TP}_{\text{embedded}} \dots [\text{AspP}_{\text{outer}} \text{ AspP}_{\text{outer}} [\text{vP}_{\text{embedded}} \text{ V-faan} \dots]]]]]]$$

We also argue that the above analysis readily extends to a scope mismatch phenomenon of aspectual suffixes — the so-called *aspect lowering* (or *experiential lowering*) — which has been analyzed by Grano (2014), N. Huang (2018), and C.-T. J. Huang (2022) as evidence for restructuring in Chinese. Moreover, we suggest that the *absence* of the lowering phenomenon of certain postverbal aspectual elements, as well as the *lack* of exceptional scopal behavior of other ‘again’-type elements like *zai* and *-gwo*, follow from (i) their association with a lower position, AspP_{inner} (within vP), and (ii) the minimal size of nonfinite clauses being vP. As later fleshed out in details, intervention by embedded AspP_{inner} is always active even for vP-sized nonfinite clauses, deriving the systematic absence of exceptional scopal behavior of these elements. Not only does the proposed view on finiteness and split aspect capture a number of intricate restructuring patterns concerning ‘again’-type and aspectual elements in two Chinese languages, but it also offers crucial support for a gradient size-based approach to finiteness of complement clauses: ranging from vP to CP, with TP marking the boundary between finite and nonfinite clauses (as in (1)).

The rest of this paper is structured as follows. Section 2 presents the Mandarin pattern regarding adverbs *you* and *zai* and their scopal behavior in nonfinite and finite clauses. Section 3 fully develops our size-based approach to finiteness, which Section 4 applies to analyze the Mandarin pattern, resorting to aspect projection licensing in connection to aspect lowering. Section 5 presents the Cantonese pattern, where parallel examples

to the Mandarin ones are found with verbal suffixes *-faan* and *-gwo*. Section 6 analyzes the Cantonese pattern, adding the split aspect approach into the general picture. Section 7 discusses our take on the minimal size of nonfinite clauses being vP and shows how this consequence can be naturally derived from our proposal. Section 8 concludes.

2 Preverbal ‘again’ skipping in Mandarin

Mandarin has two preverbal adverbs that convey the meaning of ‘again’: *you* and *zai* (a.o.: Lü 1980).⁵ Like English *again*, both *you* and *zai* have repetitive and restitutive readings (T. Xu 2012, 2016):

(10) You ‘again’

a. Repetitive

Zhe ge ren zuotian lai-guo, jintian **you** lai-le.
 this CLF person yesterday come-EXP today again come-PFV
 ‘This person came yesterday and came again today.’
 (Lü 1980: 490)

b. Restitutive

(Lisa had a bunch of red shells. Unfortunately after a while they all got very dusty and the redness faded. In need of two red shells to decorate her Christmas tree, ...)
 Ta **you** tu-hong-le qizhong liang ge beike.
 3SG again paint-red-PFV among two CLF shell.
 ‘She painted two of the shells red again.’
 (T. Xu 2012: 475)

(11) Zai ‘again’

a. Repetitive

Qu-guo-le hai keyi **zai** qu.
 go-EXP-PFV still be.permitted again go
 ‘[Even if you] have been there, [you] can still go again.’
 (Lü 1980: 499)

b. Restitutive

(The door was open before (not opened by John). Then it got closed. John wants to open the door but he was too tired, so ...)
 Ta mei **zai** ba men dakai.
 3SG not.PFV again DISP door open
 ‘He did not open the door again.’
 (T. Xu 2012: 478)

This paper concerns only the repetitive reading, but see T. Xu 2012, 2016 and Liu 2021, 2024 for extensive discussion on the restitutive reading.

⁵ Some other elements in Mandarin share a similar pronunciation with *you* and *zai* — notably, the perfective and progressive aspect markers. Unless otherwise specified, when not accompanied with a translation, *you* and *zai* are used to refer specifically to the ‘again’-type elements in the main text.

In what follows, we show that *you* exhibits a type of exceptional scopal behavior, which *zai* does not exhibit. We correlate their (dis)allowance of the scope skipping effect with their (un)embeddability under nonfinite clause-taking predicates.

2.1 Scopal behavior of *you* and *zai*

Liu (2021, 2024) and C.-T. J. Huang (2022) report that when Mandarin preverbal *you* ‘again’ appears in the matrix clause, it can take narrow scope and directly apply to the embedded predicate. Example (12) includes a context where there is an antecedent event of Xiaoming going to Hong Kong but no antecedent event of Xiangming wanting to do so. In (12a), although *you* comes before the matrix verb ‘want’, it is understood as if it is attached to the embedded predicate ‘go to Hong Kong’, resulting in a felicitous utterance in the given context. Metaphorically, *you* “skips” the main predicate and takes the embedded predicate directly as its scope just like back in (3). While previous studies have not examined other Mandarin ‘again’-type elements in relation to this skipping effect, here, we make a novel observation that *zai* does not allow the same scope skipping effect: when occurring before the matrix verb, it can only attach to the matrix verb, making (12b) infelicitous in the given context.⁶

(12) *Narrow scope context: ‘Want’ > ‘again’ > ‘go’*

(Yesterday, Xiaoming took a business trip to Hong Kong for work, despite not wanting to travel, as he was compelled by his boss. Unfortunately, he did not complete the work before returning. Today, ...)

a. *Matrix you: Scopally skipping ‘want’*

(Worried about getting fired for not finishing the work, he wants to go back to Hong Kong again to finish it.)

Xiaoming **you** xiang [qu (yi ci) Xianggang] (le).

Xiaoming again want go one time Hong.Kong PRF

Narrow scope: ‘Xiaoming wants to again go to Hong Kong (once).’

#Wide scope: ‘Xiaoming again wants to go to Hong Kong (once).’

b. *Matrix zai: Not scopally skipping ‘want’*

(It is thought that he will want to go back to Hong Kong again to finish it.)

#Xiaoming hui **zai** xiang [qu (yi ci) Xianggang].

Xiaoming will again want go one time Hong.Kong

*Narrow scope: ‘Xiaoming will want to again go to Hong Kong (once).’

#Wide scope: ‘Xiaoming will again want to go to Hong Kong (once).’

⁶ In order to retrieve the desired reading in (12a), some but not all speakers prefer adding *yi ci* ‘once’ or sentence-final perfect marker *le*. Also, (12b) contains an additional future modal *hui* ‘will’ that is absent in (12a). This is because *zai*, for reasons unknown to us, is better suited with such elements. Yet, its incompatibility with ‘again’ skipping is independent of the presence of *hui*, as demonstrated below, where *zai* still has only the matrix scope reading in the absence of *hui*:

(i) Dang women **zai** xiang [qu baoming] shi, xuexiao que shuo women cuoguo-le baoming shijian.

when 1PL again want go register time school however say 1PL miss-PFV registration time

‘When we again wanted to register, however, the school said we had missed the registration time.’

(BaoGaoSu.com n.d., accessed February 27, 2025)

Also, note that *you* cannot be embedded under *hui* due to an aspectual constraint, to be discussed in Section 2.2.

We use asterisks (*) and pound signs (#) before free translations to denote varying senses of unacceptability for the corresponding readings. An asterisk indicates that the reading is impossible for the given utterance, while a pound sign indicates that the reading is possible but infelicitous under the given context. For example, although the pound signs preceding the wide scope readings in (12) indicate their infelicity in the given context, these readings are still possible interpretations of the corresponding utterances. They can become felicitous if the context is adjusted, such as in (13).

(13) *Wide scope context: ‘Again’ > ‘want’ > ‘go’*

(Xiaoming loves traveling and is a big fan of Hong Kong. Last year, he wanted to go to Hong Kong.)

a. *Subcontext for you*

(This year, Xiaoming again wanted to go to Hong Kong.)

b. *Subcontext for zai*

(This year, it is believed that Xiaoming will again want to go to Hong Kong.)

Some other predicates like *dasuan* ‘plan’ also allow the scope skipping of *you*. In (14), which is minimally different from (12a) in that the matrix verb is replaced by ‘plan’, we again observe that *you* exhibits the scope skipping effect, whereas *zai* does not.

(14) a. *Matrix you: Scopally skipping ‘plan’*

(Worried about getting fired for not finishing the work, he plans to go back to Hong Kong again to finish it.)

Xiaoming **you** dasuan [qu (yi ci) Xianggang] (le).

Xiaoming again plan go one time Hong.Kong PRF

Narrow scope: ‘Xiaoming plans to again go to Hong Kong (once).’

#Wide scope: ‘Xiaoming again plans to go to Hong Kong (once).’

b. *Matrix zai: Not scopally skipping ‘plan’*

(It is thought that he will plan to go back to Hong Kong again to finish it.)

#Xiaoming hui **zai** dasuan [qu (yi ci) Xianggang].

Xiaoming will again plan go one time Hong.Kong

*Narrow scope: ‘Xiaoming will plan to again go to Hong Kong (once).’

#Wide scope: ‘Xiaoming will again plan to go to Hong Kong (once).’

The agent of the embedded event over which ‘again’ takes scope does not need to be the subject of the matrix predicate. Along with the internet example in (3) in Section 1, we include (116) below, where the matrix verb is an object control verb *quan* ‘urge’.

(15) *Matrix you: Scopally skipping ‘urge’*

(Wo zhiqian mai-guo baojianpin le.) Ni bu yao **you** quan wo mai!

1SG before buy-EXP health.product PRF 2SG not IMP again urge 1SG buy

‘(I have bought health products before.) Don’t urge me to buy again!’

This utterance remains perfectly felicitous even if the addressee has never urged the speaker to buy health products. Putting everything together, (12–116) lead us to the generalization in (16).

(16) *Generalization I*

The exceptional scopal behavior of ‘again’ is found with *you* but not with *zai*.

The availability of this scope skipping effect observed with *you* depends on the choice of matrix verbs: only nonfinite clause-taking predicates allow for *you* to “skip” scope. In (17), we have a context that does not convey an antecedent event of believing but conveys an antecedent event of appearing. This context is compatible with a target sentence where the matrix verb is ‘believe’ and scopes over *you*. In this context, *you* preceding matrix ‘believe’ is infelicitous in (17a), indicating that *you* cannot “skip” ‘believe’, in contrast to (17b), where *you* surfacing in the embedded clause can produce the intended interpretation.

(17) *Narrow scope context: ‘Believe’ > ‘again’ > ‘appear’*

(Initially, during the 2003 SARS outbreak, Xiaoming did not believe in the existence of coronavirus, thinking that it was just a flu. After the outbreak, there were no reported cases of coronavirus until the COVID-19 outbreak in 2019. Miraculously, by that time, Xiaoming had become an epidemiologist and collected a sample containing the COVID-19 virus, which led him to come to believe that coronavirus had appeared before, and it had appeared again.)

(Notationally: Xiaoming’s belief state in 2003 = {✗coronavirus in 2003};

Xiaoming’s belief state in 2019 = {✓coronavirus in 2003, ✓coronavirus in 2019}.)

a. *Matrix you: #*

#Xiaoming **you** xiangxin [chuxian-le guanzhuangbingdu].

Xiaoming again believe appear-PFV coronavirus

*Narrow scope: ‘Xiaoming believes that coronavirus again appeared.’

#Wide scope: ‘Xiaoming again believes that coronavirus appeared.’

b. *Embedded you: OK*

Xiaoming xiangxin [**you** chuxian-le guanzhuangbingdu].

Xiaoming believe again appear-PFV coronavirus

Narrow scope: ‘Xiaoming believes that coronavirus again appeared.’

In Mandarin, ‘believe’ takes finite clause complements, which are standardly assumed to be CPs (C.-T. J. Huang 2022), while ‘want’ takes nonfinite clause complements (e.g., TPs or vPs). An extensive examination by Liu (2021, 2024) shows that the narrow scope interpretation of *you* is allowed only with the nonfinite clause-taking predicates listed in (18) but not with the finite clause-taking predicates listed in (19). This observation suggests that *you* cannot “skip” across a finite clause boundary.

(18) *Nonfinite clause-taking predicates: ✓You skipping*

bi(po) ‘force’, *changshi* ‘try’, *dasuan* ‘plan’, *jihua* ‘plan’, *jueding* ‘decide’, *mingling* ‘order’, *quan(shuo)* ‘urge’, *rang* ‘let’, *shefa* ‘try’, *tingzhi* ‘stop’, *tiyi* ‘propose’, *tuijian* ‘recommend’, *weixie* ‘threaten’, *xiang(yao)* ‘want’, *xuanze* ‘choose’, *zhunbei* ‘prepare’, etc.

(Liu 2024: 215, adapted)

- (19) *Finite clause-taking predicates: ✗You skipping*
faxian ‘discover’, *fouren* ‘deny’, *gaosu* ‘tell’, *huaiyi* ‘suspect’, *queren* ‘confirm’, *renwei* ‘think’, *sheng-ming* ‘declare’, *shuo* ‘say’, *xiangxin* ‘believe’, *zhidao* ‘know’, etc.
 (Liu 2024: 215)

Hence, we generalize in (20) that the scope skipping of *you* is limited to nonfinite clause-taking predicates.⁷

(20) *Generalization II*

The exceptional scopal behavior of ‘again’ can cross **nonfinite** but not **finite** clause boundaries.

We wrap up this subsection by showing an example with multiple nonfinite clause boundaries “skipped” across. In (21), the matrix *you* skips over the matrix *shefa* ‘try’ and middle *rang* ‘let’, and is interpreted as modifying the innermost embedded predicate *xiangxin* ‘believe’.⁸

- (21) (Xiaoming is a good friend of mine. He knows that I had been a true believer of love, until I recently broke up with my partner. Trying to cheer me up, he recommended a famous romance TV series to me, hoping that it will make me believe in love again. He has never done anything like this before.)

Xiaoming **you** shefa [rang wo [xiangxin aiqing]] (le).

Xiaoming again try let 1sg believe love PRF

Narrow scope: ‘Xiaoming tried to let me again believe in love.’

#Intermediate scope: ‘Xiaoming tried to again let me believe in love.’

#Wide scope: ‘Xiaoming again tried to let me believe in love.’

2.2 Structural differences between *you* and *zai*

A contrast between *you* and *zai*, observed by Lin & Liu (2009), is their embeddability by certain nonfinite clause-taking predicates. While *you* cannot surface in a nonfinite clause headed by verbs like *keyi* ‘be permitted’ and *xiang* ‘want’, as in (22), *zai* can, as in (23).

- (22) a. *Xiaoming keyi [you lai] le.
 Xiaoming be.permitted again come PRF
 Intended: ‘Xiaoming has been permitted to again come.’
 (Lin & Liu 2009: 1186, adapted)

⁷ It does not mean that all nonfinite clause-taking predicates allow *you* skipping. As we will see in Section 4, some nonfinite clause-taking predicates may take two clause sizes, TP and vP, and it is only vP that allows *you* skipping. Some predicates may even only take TP, but not vP, and disallow *you* skipping altogether (see Appendix 1).

⁸ Again, the infelicity of the intermediate and wide scope readings does not disqualify them as possible interpretations. Contexts where they become felicitous are provided below:

- (i) a. *Intermediate scope context: ‘Try’ > ‘again’ > ‘let’ > ‘believe’*
 (Xiaoming is a good friend of mine. Before we met, he and his girlfriend were a legendary couple at our school, known for their touching love story. It was so moving that it made me believe in love. I haven’t been as lucky: I never found my true love and eventually lost faith in it. That changed when Xiaoming and I became friends. He shared more stories about his relationship, and through them, he’s been trying to restore my belief in love.)
 b. *Wide scope context: ‘Again’ > ‘try’ > ‘let’ > ‘believe’*
 (I despise love and find it impossible to believe in it, but my friend Xiaoming is determined to change my mind. He never stops trying!)

- b. *Xiaoming xiang [**you** qu Xianggang].
 Xiaoming want again go Hong.Kong
 Intended: ‘Xiaoming wants to again go to Hong Kong.’
- (23) a. Xiaoming keyi [**zai** qu Taipei].
 Xiaoming be.permitted again go Taipei
 ‘Xiaoming is permitted to again go to Taipei.’
 (Lin & Liu 2009: 1195)
- b. Xiaoming xiang [**zai** qu Xianggang].
 Xiaoming want again go Hong.Kong
 ‘Xiaoming wants to again go to Hong Kong.’

Lin & Liu (2009) accounts for this contrast using the *dynamic* ([+D]) and *static* ([−D]) aspectual features, originally proposed by Shen (2004). (See also Lin 2011.) A [+D] Asp⁰ requires that the predicate be an accomplishment, an achievement, or an activity under the Vendlerian classification (Vendler 1957). Lin & Liu (2009) propose that *you* must be licensed by the closest Asp⁰ head that carries [+D]. According to them, *keyi* ‘be permitted’ and *xiang* ‘want’ in (22) take a complement that is incompatible with a [+D] Asp⁰, necessitating a static meaning. This is evidenced in (24) by the failure to embed perfective *-le*, which can only occur with a dynamic predicate (Shen 2004: 175). Since the complement clauses in (24) are not [+D], *you* embedding is not licensed.

- (24) *Xiaoming keyi /xiang [qu-le Xianggang].
 Xiaoming be.permitted want go-PFV Hong.Kong
 Intended: ‘Xiaoming is permitted/wants to have gone to Hong Kong.’

We observe that this is not the case for *rang* ‘let’, which also takes nonfinite clauses as complements but can project a [+D] Asp⁰ in its complement, as shown in (25) with *-le*.⁹

⁹ Anticipating an extensive discussion about aspect lowering in Section 3.1, we briefly note here that (25) is ambiguous between a reading where the perfective aspect is interpreted locally in the embedded clause and a reading where it is taking scope over the matrix. What is crucial here is that the latter reading is possible.

We would also like to highlight that both (26) and (27) do not involve the perfective aspect taking scope over the matrix predicate. We can see this from the example below (using *you* for demonstration), which cannot be followed up with a negative statement regarding the embedded content (i.e., ‘but I still do not believe in love’) without yielding a contradiction:

- (i) Zhe bu ju rang wo [**you** xiangxin-le aiqing]. (#Danshi wo haishi bu xiangxin aiqing.)
 this CLF series let 1SG again believe-PFV love but 1SG still not believe love
 ‘This series let me again believe in love. (#But I still do not believe in love.)’

We take this observation as key evidence that the core contrast between (24) and (25) (without aspect lowering) lies in whether a predicate can select for a complement aspectually marked as [+D] (rather than, for example, in whether the matrix predicates themselves are marked with [+D]). Moreover, while we attribute the impossibility of aspect lowering in (26) to the presence of *you* in the embedded clause, which renders the clause too large for aspect lowering to take place—a point we elaborate on in Section 4.3—we acknowledge that the impossibility of aspect lowering in (27) remains unexplained, although the cause appears to be independent and perhaps idiosyncratic.

- (25) Zhe bu ju rang wo [xiangxin-le aiqing].
 this CLF series let 1SG believe-PFV love
 ‘This series let me believe in love.’

As expected, *rang* can embed *you* as in (26). On the other hand, *zai* is not restricted to dynamic aspects and can be embedded by ‘be permitted’ or ‘want’ as in (23) above, and so can it be embedded under ‘let’ as in (27) below. Note that *rang* is compatible with [+D] but it does not require [+D] in its complement, as can be seen from the absence of *-le* in (21) above. For more discussion, see Section 4.3.

- (26) Zhe bu ju rang wo [**you** xiangxin-le aiqing].
 this CLF series let 1SG again believe-PFV love
 ‘This series let me again believe in love.’
- (27) Zhe bu ju rang wo [**zai** (yi ci) xiangxin-le aiqing].
 this CLF series let 1SG again one time believe-PFV love
 ‘This series let me again believe in love.’

To complete the picture, (28) (reproduced from (17b)) and (29) show that both *you* and *zai* are embeddable by finite clause-taking predicates, such as *xiangxin* ‘believe’.

- (28) Xiaoming xiangxin [**you** chuxian-le guanzhuangbingdu].
 Xiaoming believe again appear-PFV coronavirus
 ‘Xiaoming believes that there again appeared coronavirus.’
- (29) Xiaoming xiangxin [hui **zai** chuxian guanzhuangbingdu].
 Xiaoming believe will again appear coronavirus
 ‘Xiaoming believes that there again appeared coronavirus.’

We have examined the nonfinite clause-taking predicates in (18) that resist a dynamic aspect independently, and they all resist embedding *you* as well. On the other hand, all the finite clause-taking predicates in (18) allow for a dynamic aspect and may embed *you*. The data are summarized in Table 2, and the compatibility with the scope skipping effect of ‘again’-type elements and their embeddability are correlated as in (30).

(30) *Correlation I*

In Mandarin, an ‘again’-type element exhibits exceptional scopal behavior *iff* it cannot surface in an embedded nonfinite clause without a dynamic aspect.

Another contrast between *you* and *zai* is whether they are structurally higher than aspectual elements. Only *you* but not *zai* can be used with experiential aspect marked by the suffix *-guo*, as in (31).¹⁰

¹⁰ The unacceptability of (31b) does not stem from a clash between *-guo* and the future modal *hui*, as they can co-occur:

- (i) Wo mingnian zhe ge shihou jiu **hui** yijing qu-guo Xianggang le.
 1SG next.year this CLF time then will already go-EXP Hong.Kong PRF
 ‘I will have already gone to Hong Kong by this time next year.’

Table 2 Clause-taking predicates and their compatibility with dynamic aspect and embedded *you*

Predicates	Finiteness	Dynamic aspect	Embedded <i>you</i>
a. <i>dasuan/jihua</i> ‘plan’, <i>jueding</i> ‘decide’, <i>tingzhi</i> ‘stop’, <i>tiyi</i> ‘propose’, <i>tuijian</i> ‘recommend’, <i>weixie</i> ‘threaten’, <i>xiang(yao)</i> ‘want’, <i>zhunbei</i> ‘prepare’	nonfinite	✗	✗
b. <i>bi(po)</i> ‘force’, <i>changshi/shefa</i> ‘try’, <i>mingling</i> ‘order’, <i>quan(shuo)</i> ‘urge’, <i>rang</i> ‘let’, <i>xuanze</i> ‘choose’	nonfinite	✓	✓
c. <i>faxian</i> ‘discover’, <i>fouren</i> ‘deny’, <i>gaosu</i> ‘tell’, <i>huaiyi</i> ‘suspect’, <i>queren</i> ‘confirm’, <i>renwei</i> ‘think’, <i>shengming</i> ‘declare’, <i>shuo</i> ‘say’, <i>xiangxin</i> ‘believe’, <i>zhidao</i> ‘know’	finite	✓	✓

(31) *Structural height with the experiential suffix -guo*

- a. Wo jinnian **you** qu-guo (yi ci) Xianggang.
1SG this.year again go-EXP one time Hong.Kong
‘I have gone to Hong Kong (once) again this year.’
- b. *Wo mingnian hui **zai** qu-guo (yi ci) Xianggang.
1SG next.year will again go-EXP one time Hong.Kong
Intended: ‘I will have gone to Hong Kong (once) again next year.’

Similar contrasts are found with the perfective adverb *yijing* ‘already’ collocated with the perfective suffix *-le* (not to be confused with sentence-final perfect *le*; cf. [Erlewine 2017](#)), as in (32).

(32) *Structural height with the perfective adverb yijing*

- a. Ta **you** yijing shiqu-le zongying.
3SG again already lose-PFV trace
‘His trace is again already lost.’
([Facebook 2017](#))
- b. *Ta **zai** yijing shiqu-le zongying.
3SG again already lose-PFV trace
Intended: ‘His trace again already disappeared.’

For *zai* to co-occur with *yijing*, it must follow *yijing*:

- (33) Danshi wo yijing **zai** zhao-le nanpengyou le.
but 3SG already again find-PFV boy.friend PRF
‘But I have already found a boyfriend again.’
([Baidu 2016](#))

This pattern extends to the progressive marker (*zheng*)*zai*. You can precede the progressive marker in (34a), whereas *zai* ‘again’ cannot in (34b).

(34) *Structural height with the progressive prefix (zheng)zai*

- a. Wo jia de mao **you** (zheng)zai gan guai shi le.
 1SG house POSS cat again PROG do weird thing PRF
 ‘The cat in my house is doing weird things again.’
 (Books.com.tw 2017, adapted)
- b. Wo jia de mao hui **zai** (*zai /*zhengzai) gan guai shi.
 1SG house POSS cat will again PROG PROG do weird thing
 Intended: ‘The cat in my house will be doing weird things again.’

Again, it is possible to place *zai* ‘again’ after (zheng)zai. In contrast, you cannot follow (zheng)zai:

- (35) Women zhengzai {**zai**/ ***you**} zou Riben ren de laolu me?
 1PL PROG again again walk Japan person POSS old.path Q
 ‘Are we again following the old path of the Japanese?’
 (Obj.cc 2010, adapted)

The correlation observed in (31–34) is generalized in (36).¹¹

(36) *Correlation II*

An ‘again’-type element exhibits exceptional scopal behavior *iff* it is structurally higher than aspectual elements.

Further, *you* and *zai* may co-occur, but in a fixed order — only *you zai* but not *zai you* is acceptable, as in (37).¹² In other words, *you* is syntactically higher than *zai*. This observation is consistent with the above two correlations.

- (37) a. **You zai** shang-le laotou yi wan jiu.
 again again give-PFV old.man one CLF wine
 ‘(He) again gave the old man a bowl of wine.’
 (Ta Kung Pao 2025)
- b. ***Zai you** shang-le laotou yi wan jiu.
 again again give-PFV old.man one CLF wine

¹¹ An apparent counterexample for (36) is that *zai* ‘again’ may precede negation like *bu* or *meiyou*, and the resulting expression *zai bu/meiyou* conveys ‘never do/did again’ rather than ‘again do/did not’, displaying an apparent scope mismatch. We suggest that these expressions, as pointed out by an anonymous reviewer, are contractions of *zai ye bu* and *zai ye meiyou*, which we further speculate to be derived from the unconditional use of *zai* rather than its repetitive use. Below is an illustrative example of its unconditional use:

- (i) Wo zai zenme nuli, ye zao bu chu bie de juzi le.
 1SG again how make.effort also make not out other MOD sentence PRF
 ‘No matter how much effort I make, I cannot come up with any other sentence’
 (Reddit 2021)

Hence, the derivational or diachronic path we propose in a nutshell is that the expression begins its life with *zai zenme yang ye bu/meiyou* ‘do/did not no matter what’, which shortens to *zai ye bu/meiyou* and then further to *zai bu/meiyou*, yielding an unconditional reading that resembles a reading in which ‘again’ scopes below negation. The apparent scope mismatch is therefore not a true counterexample for (36), as the source of the reading is not the repetitive *zai* but the unconditional *zai*.

¹² Curiously, the meaning of *you zai* does not add up to indicate the action to occur for the third/fourth time (i.e., ≠ reoccur again) but simply the second time. Without going into details, we suggest that it is due to the presuppositional nature of ‘again’ and that *you*’s semantic input is only the truth-conditional content without *zai*’s presupposition.

3 A size-based approach to finiteness

Before we provide an analysis of the exceptional scopal behavior of *you* ‘again’, we first lay out our size-based approach to finiteness. We take an excursion to highlight how *aspect lowering*, coupled with other cross-clausal dependencies, motivates a hierarchical, gradient notion of finiteness (with terms adopted from C.-T. J. Huang 2022: 22):

- (38) a. CPs are “the most finite” and the most opaque domain.
 b. TP_s are “somewhat infinitival” and a partially transparent domain.
 c. VP_s are “the most infinitival” and the most transparent domain.

We clarify and formulate the syntactic nature of this conception below.

3.1 Aspect lowering as diagnosis of nonfinite clause sizes

One of the earliest arguments for finiteness distinction in Chinese comes from aspect lowering (Y.-H. A. Li 1985: 35ff, C.-T. J. Huang 1989: 190ff, Y.-H. A. Li 1990: 18ff). It refers to the scope mismatch where aspectual suffixes like the experiential suffix *-guo*, when appearing in a nonfinite clause, take scope over the matrix predicate (Grano 2014, N. Huang 2018, C.-T. J. Huang 2022). Some examples are given in (39). The interpretation is similar to placing *-guo* on the matrix predicate, for instance with *quan* ‘urge’ in (40).¹³

(39) *Nonfinite clause-taking predicate: High interpretation of -guo*

- a. Lisi shefa [xiuli-guo zhe tai jiqi].
 Lisi try repair-EXP this CLF machine
 *Narrow scope: ‘Lisi tries to have repaired this machine.’
 Wide scope: ‘Lisi has tried to repair this machine.’
 (N. Huang 2018: 351)
- b. Zhangsan (congqian) quan Lisi [jie-guo yan].
 Zhangsan before urge Lisi quit-EXP cigarette
 *Narrow scope: ‘Zhangsan urged Lisi to have quit smoking (before).’
 Wide scope: ‘Zhangsan has (before) urged Lisi to quit smoking.’
 (C.-T. J. Huang 2022: 32, with ‘before’ made optional)
- c. Zhangsan (congqian) bi Lisi [xuan-guo weijifen].
 Zhangsan before force Lisi choose-EXP calculus
 *Narrow scope: ‘Zhangsan has forced Lisi to have chosen calculus (before).’
 Wide scope: ‘Zhangsan has (before) forced Lisi to choose calculus.’

- (40) Zhangsan (congqian) quan-guo Lisi [jie yan].
 Zhangsan before urge-EXP Lisi quit cigarette
 *Narrow scope: ‘Zhangsan has urged Lisi to have quit smoking (before).’
 Wide scope: ‘Zhangsan has (before) urged Lisi to quit smoking.’

¹³ For independent reasons, *-guo* cannot attach to *shefa* ‘try’. However, the scope mismatch is still possible in (39a), suggesting that there is no correlation between aspect lowering and a matrix predicate’s compatibility with *-guo*.

The high interpretation of *-guo* above indicates a lack of actuality entailment of the embedded event. For example, the felicity of (39b) does not require that Lisi accepted Zhangsan's suggestion and actually quit smoking. This obligatory wide scope can be made even more explicit by adding an aspectual marker like the progressive (*zheng*)*zai* in the matrix, which results in a conflict in aspectual meanings:

(41) *Matrix progressive + embedded experiential* $\rightsquigarrow$ *matrix aspect clash*

*Zhangsan (*zheng*)*zai* quan Lisi [*jie-guo* yan].

Zhangsan PROG urge Lisi quit-EXP cigarette

Intended: 'Zhangsan is urging Lisi to have quit smoking.' or 'Zhangsan has been urging Lisi to quit smoking.'

For this reason, we take the lack of actuality entailment to be a signal of aspect lowering — that is, an embedded *-guo* being interpreted with respect to the matrix aspect. Put differently, aspect lowering is observed only when the event denoted by the embedded predicate is not necessitated to happen.¹⁴ The above scopal mismatch is not found when the aspectual suffix is embedded in a finite clause:

(42) *Finite clause-taking predicate: Low interpretation of -guo*

Zhangsan shuo/xiangxin [Lisi (*congqian*) *jie-guo* yan].

Zhangsan say believe Lisi before quit-EXP cigarette

Narrow scope: 'Zhangsan says/believes that Lisi has quit smoking (before).'

*Wide scope: 'Zhangsan has said/believed that Lisi quits smoking (before).'

(C.-T. J. Huang 2022: 32, adapted)

The (un)availability of aspect lowering determines whether the embedded subject must be covert, as illustrated by the contrast between (39) and (42), and thus serves as a reliable diagnosis for finiteness. Early proposals in works like Y.-H. A. Li 1985, 1990, and C.-T. J. Huang 1989 suggest that aspect lowering reflects the lack of *Aux*⁰ or *Infl*⁰ in nonfinite clauses, alluding to the idea that the finiteness distinction is manifested as the presence or absence of designated finite heads/features (e.g., [$\pm$ FINITE] or [$\pm$ TENSE] in Lin 2011 and Zhang 2019).

However, not all nonfinite clauses allow aspect lowering, as insightfully pointed out by N. Huang (2018) and followed up by C.-T. J. Huang (2022). With the presence of (i) the semicomplementizer *shuo*, (ii) the adverb *ye* 'also', or (iii) internal foci, aspect lowering becomes unavailable. In (43), the embedded *-guo* cannot be interpreted with respect to the matrix predicates.

(43) a. *Semicomplementizer shuo blocking aspect lowering*

*Lisi cengjing shefa [*shuo* *zuo-guo* zhe dao cai].

Lisi previously try COMP make-EXP this CLF dish

Intended: 'Lisi has previously tried to cook this dish.'

(C.-T. J. Huang 2022: 49)

¹⁴ Grano (2014: fn. 1) mentions that for some speakers, an actuality entailment is required when *-guo* is embedded in nonfinite clauses — that is, both matrix and embedded events are realized. We, as well as our Mandarin consultants, do not share this requirement. See also Grano 2019 for experimental evidence that no actuality entailment of the embedded event is required.

b. *Adverb ye ‘also’ blocking aspect lowering*

*Lisi cengjing shefa [ye zuo-guo zhe dao cai].
 Lisi previously try also make-EXP this CLF dish
 Intended: ‘Lisi has previously tried to also cook this dish.’
 (C.-T. J. Huang 2022: 50)

c. *Internal focus blocking aspect lowering*

*Zhangsan bi Lisi [lian weijifen dou xuan-guo].
 Zhangsan force Lisi even calculus all choose-EXP
 Intended: ‘Zhangsan has forced Lisi to choose even calculus.’

We know that these clauses are still nonfinite since overt subjects and future modal *hui* ‘will’ are disallowed (C.-T. J. Huang 2022). As argued by N. Huang (2018: 352ff), these elements (i–iii) independently signal (at least) the presence of TP layers, as they occur above the future tense marker *jiang* (argued to be a syntactic tense by N. Huang (2015)), the future modal *hui* ‘will’, and the progressive aspect (*zheng*)/*zai* in root clauses.¹⁵ The empirical takeaway is that some nonfinite clauses are structurally *larger* than some others and block aspect lowering.¹⁶ Therefore, a simple dichotomy of [$\pm$ FINITE] features is insufficient to account for the finiteness distinction in Chinese.

3.2 Finiteness as a hierarchy of clause sizes

To capture the distribution of aspect lowering, we follow He (2020, 2024) and C.-T. J. Huang (2022) and propose that there are at least three types of complement clauses, as in (44):

¹⁵ The status of the semicomplementizer *shuo* is subject to debate and will be picked up in Section 3.2.

¹⁶ C.-T. J. Huang (2022) suggests that internal topicalization also signals larger nonfinite clauses, demonstrated with the following example, where the object is topicalized in a nonfinite clause:

- (i) *Wo changshi [zhe zhong shousi chi-guo t].
 1SG try this kind sushi eat-EXP
 Intended: ‘I have tried to eat this kind of sushi.’
 (C.-T. J. Huang 2022: 50)

We agree with C.-T. J. Huang and find this example to be unacceptable under both high and low interpretations of *-guo*. However, we notice that high attachment of *-guo* is also unacceptable in this case, and the unacceptability does not arise from any semantic clash between *changshi* and *-guo*:

- (ii) a. *Wo changshi-guo [zhe zhong shousi chi t].
 1SG try-EXP this kind sushi eat
 Intended: ‘I have tried to eat this kind of sushi.’
 b. Wo changshi-guo [chi zhe zhong shousi].
 1SG try-EXP eat this kind sushi
 ‘I have tried to eat this kind of sushi.’

Prompted by this observation, we follow Chen & Yip (to appear) and claim that the above examples are deviant due to an independent violation of antilocality of the internal topic movement itself. Evidence includes the observation that (i) and (iia) can be rescued by inserting a manner adverb in the embedded clause. In this case, aspect lowering is possible with internal topicalization. Note that the ability to accommodate internal topicalization has often been taken to signal finiteness (T.-C. Tang 2000, Paul 2002, 2005, Grano 2017), syntactic tense (Lin 2011, 2015), or at the very least, a larger clause size like TP (N. Huang 2018, C.-T. J. Huang 2022). Chen & Yip, however, demonstrate that even very small nonfinite clauses (e.g., vP) can still embed internal topics, which are predicted to allow aspect lowering in our account. Interested readers are referred to their work for more discussion.

(44) *Verb complement clause sizes*

- | | |
|--|--|
| a. V [CP ... [TP ... [AspP ... [vP ... [VP ...]]]]] | type I (finite) |
| b. V [TP ... [AspP ... [vP ... [VP ...]]]] | type II (nonfinite without Aspect restructuring) |
| c. V [vP ... [VP ...]] | type III (nonfinite with Aspect restructuring) |

Rather than relying on a $[\pm\text{FINITE}]$ feature on T^0 , this approach attributes finiteness distinction to clause sizes, which are a result of structure building.¹⁷ In a nutshell, we propose that whether AspP is projected (i.e., type II vs. III) determines the availability of aspect lowering (as well as ‘again’ skipping in Section 4), and whether CP is projected (i.e., type I vs. II) determines the availability of other cross-clausal dependencies. Note that this view is not necessarily incompatible with the designated feature approach: N. Huang (2018: 367–368), in addition to size differences, suggests that *shuo* may be ambiguous between a finite complementizer and a nonfinite one, resulting in nonfinite CPs. (See also Huang & Cai 2025 for similar suggestions on Teochew complementizer *tan*.) Moreover, Case licensing of overt subjects, as traditionally assumed (Chomsky 1980, 1981, Y.-H. A. Li 1985, 1990), results from a specification on T^0 . We may imagine a hybrid approach where both size and the $[\pm\text{FINITE}]$ feature play a role. Nevertheless, recasting such analytic components in the size-based approach is possible. C.-T. J. Huang (2022) proposes that the CP projected by *shuo* is structurally removed/“Exfoliated” in nonfinite clauses (an issue we will return to later). Moreover, he attributes the temporal independence of finite clauses to a null operator in SpecCP that A’-binds T. This is reminiscent to Chomsky’s (2008) *feature inheritance*, where T^0 obtains its full-fledged ϕ -features from the phase head, C^0 , for agreement and Case licensing. Here, we adopt a size-based view rather than a hybrid one, not only for maintaining theoretical uniformity, but also for empirical concerns over the locality profiles of cross-clausal dependencies that are best accounted for by size differences, as discussed below.¹⁸

Our “gradient” clause size approach differs from the bipartite size approach by, for example, Grano (2012, 2014, 2015, 2017), where complement clauses can only be either CP/TopicP or vP, in that finiteness should not be understood as an all-or-nothing property but a gradient hierarchical notion. To elaborate, we follow Cinque’s (1999) functional sequencing and assume that functional heads like T^0 and Asp^0 are always projected and each selected by a higher functional head. Thus, since the presence of CP entails the presence of TP and AspP, dependencies that cannot cross type II clauses (e.g., aspect lowering) cannot cross type I clauses, either. Put differently, if type II clauses constitute a transparent domain for a certain dependency, the smaller type III clauses must also be a transparent domain for it, but the larger type I clauses are not necessarily so (e.g., internal topicalization, to be discussed later in this section). This derived consequence suggests a three-tiered implicational hierarchy, fully conforming with Wurmbrand & Lohninger’s (2023) ICH in holding a unidirectional view of clausal transparency among three types of complements defined semantically, as in (2), copied below in (45).

¹⁷ In other words, we adopt a “reductionist” view in abandoning finiteness as a primitive notion in the grammar as the $[\pm\text{FINITE}]$ feature. The so-called finiteness distinction is a cluster of properties associated with clauses of different sizes, at least in Chinese (resonating Grano 2017).

¹⁸ While we focus primarily on cross-clausal dependencies in this paper, other traditional hallmarks of finiteness can also be captured under the size-based view, such as the ban on overt subjects in nonfinite clauses. Assuming that overt subjects require Case licensing by a T with $[\mu\text{Case:NOM}]$ (or $[\mu\phi]$ if one assumes that Case assignment is a by-product of ϕ -agreement), and further that $[\mu\text{Case:NOM}]$ originates on the phase head C, which is then “donated” to T by feature inheritance (Chomsky 2008), it follows that clauses smaller than CP cannot have an overt subject (but either a PRO in control or a trace in raising). Effectively, only CP (Type I) has overt subjects but not TP (Type II) and vP (Type III), deriving this traditional hallmark of finiteness. We thank an anonymous reviewer for raising this concern.

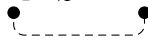
(45) *Implicational complementation hierarchy*

MOST INDEPENDENT		LEAST INDEPENDENT
LEAST TRANSPARENT	proposition > situation > event	MOST TRANSPARENT
LEAST INTEGRATED		MOST INTEGRATED

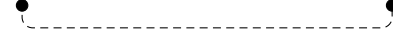
(Wurmbrand & Lohninger 2023: 188)

Their model assumes that each complement has a “minimal size” to be mapped onto — CP for “proposition,” TP (or Wurmbrand’s (2014b) wollP) for “situation,” and vP for “event” — and the unidirectional implication is obtained via syntactic containment. This view predicts three distinct groups of dependencies, one that distinguishes type III from type II, one that distinguishes type II from type I, and one that is insensitive to clause sizes. They also suggest that the mapping can be imperfect to a limited extent for capturing variation, especially when no grouping of the dependencies is found: an “event” complement can be mapped onto a larger size like TP or even CP, but a “proposition” complement can never be mapped onto a smaller size like vP or TP. If a language uniformly maps “situation” to CP (or “event” to TP), there would be only two groups of dependencies. Empirically, Chinese shows three groups of dependencies, as we will show below. Our proposal in (44) is defined *syntactically*. Thus, if an “event” complement projects a TP (our type II), dependencies that can only cross vP would be blocked.

We now discuss how aspect lowering supports the above view. To begin with, recent accounts (Grano 2014, N. Huang 2018, C.-T. J. Huang 2022) assume a lexicalist view on verb suffixation (Gu 1993, 1995, Huang, Li & Li 2009, S.-W. Tang 2010, 2015; cf. Chomsky 1970, 1993). The lexicalist approach posits that verbal suffixes are base-generated with the verbal root as a lexical V^0 in the syntax, rather than projecting a functional head that attracts overt head movement.¹⁹ The suffixed verbs need to be licensed by an Asp^0 , which may be achieved by covert LF raising (e.g., Gu 1995) or agreement (e.g., Grano 2014). We adopt the second option, as schematized in (46).

- (46) $[_{AspP} Asp [_{vP} V\text{-suffix} \dots]$ *licensing suffixes by aspectual agreement*
- 

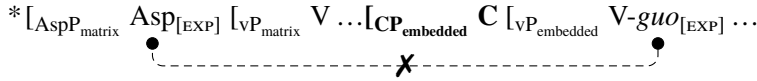
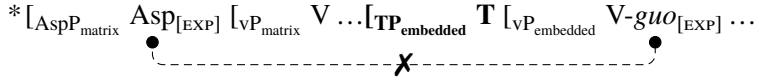
Following this lexicalist view, aspect lowering is analyzed as an instance of aspect agreement across (non-finite) clause boundaries. Adopting the assumption that Mandarin nonfinite clauses may lack a local $AspP$ and be as small as vPs (i.e., type III), an aspectual suffix embedded in a nonfinite clause has to agree with matrix Asp^0 , thus acquiring matrix scope (Grano 2014, N. Huang 2018, C.-T. J. Huang 2022; cf. Liu 2021, 2024), as illustrated in (47) with the experiential aspect.²⁰

- (47) $[_{AspP_{matrix}} Asp_{[EXP]} [_{vP_{matrix}} V \dots [_{vP_{embedded}} V\text{-}guo_{[EXP]} \dots]]$
- 
- (N. Huang 2018: 360, with [EXP] added)

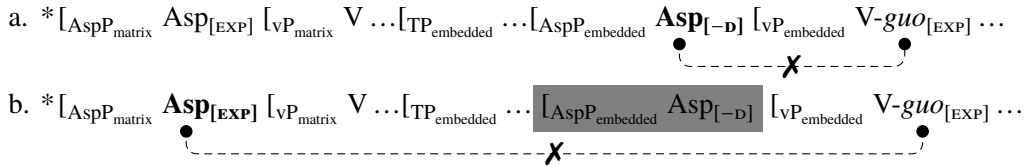
The data we have seen above suggest that when the embedded clause is so large that it encompasses a C^0 or even just T^0 , an aspectual suffix cannot agree with matrix Asp^0 , which either yields a low interpretation or ungrammaticality:

¹⁹ For the alternative nonlexicalist view in Chinese languages, see S.-W. Tang 1998, 2003, 2006, Tsai 2001, 2008, Chan, Lee & Yip 2022, Lee & Yip 2024b.

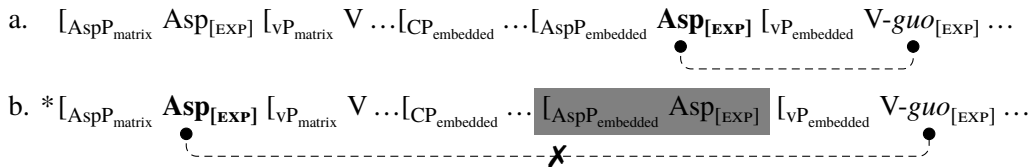
²⁰ While we follow previous works on aspect lowering in assuming the lexicalist approach, we note that it is not impossible to implement aspect lowering under a nonlexicalist approach, as long as suffixes themselves do not carry the aspectual semantics. For example, one may assume that verbal suffixes project $AspP$ in the embedded clause, and agree with an aspectual operator in the specifier position of the matrix $AspP$. We leave the comparison between these two approaches to future work.

(48) a. *Finite clauses like (42)*b. *Nonfinite clauses like (43)*

While N. Huang (2018: 359ff) and C.-T. J. Huang (2022: 47–52) derive the impossibility of matrix aspect agreement in these constructions from the phasal head status of C⁰ and from the intervening temporal features of T⁰, we offer a unified analysis, categorizing all these cases under (defective) intervention effects (Chomsky 2000). For nonfinite clause-taking predicates, they are semantically incompatible with the experiential aspect in embedded clauses. Hence, the embedded Asp⁰ does not come with the experiential [EXP] features (but with other features like static aspect [–D], see Section 4.1), and cannot license the embedded -*guo* which requires [EXP], as schematized in (49a). Nevertheless, it is also impossible for -*guo* to agree with the matrix Asp⁰ that has [EXP], due to the intervention by the embedded Asp⁰ as in (49b). According to Chomsky (2000), even elements partially matching in features induce defective intervention effects. (See also Rizzi 2001 for a similar effect implemented under Relativized Minimality with *superfeatures*.) As a result, neither aspect lowering nor local licensing is allowed with TP-sized nonfinite clauses, leading to ungrammaticality.

(49) *Intervention effects with nonfinite TPs*

Of course, under finite clauses taken by predicates like *shuo* ‘say’, the embedded Asp⁰ may have [EXP]. The embedded -*guo* must then agree with the closest Asp⁰, the embedded one rather than the matrix one, as required by the locality constraint on Agree (e.g., Chomsky 2000). Thus, the embedded -*guo* is still licensed and yields a low interpretation, only that aspect lowering is impossible due to (standard) intervention effects, as illustrated in (50).

(50) *Intervention effects with finite CPs*

After seeing how AspP determines the locality profiles of type II and type III clauses with regard to aspect lowering, let us consider the difference between type I and type II clauses in the case of topicalization. As argued by N. Huang (2018), movement of clause-internal topics below subjects, while also bounded by finite CPs as in (51a), may cross a nonfinite TP boundary, whose presence is signaled by the semicomplementizer *shuo* in (51b). (See also Shyu 1995, C.-T. J. Huang 2022, Chen 2023, to appear, and references therein.)

(51) *Internal topicalization crossing nonfinite TPs but not finite CPs*

- a. *Wo [zhe pian baogao] xiangxin [_{CP} Lisi xie-wan-le t].
 1SG this CLF report believe Lisi write-finish-PFV
 Intended: ‘I believe that Lisi has already written this report.’
 (N. Huang 2018: 351)
- b. Lisi [zhe pian baogao] hui shefa [_{TP} (shuo) zai zhe zhou nei xie-wan t].
 Lisi this CLF report will try COMP at this week in write-finish
 ‘Lisi will try to finish writing this report this week.’
 (N. Huang 2018: 359)

On the other hand, external topicalization, with a landing site above subjects, may freely cross a finite CP boundary, as in (52a). Also, as expected, it may cross nonfinite TPs, as in (52b).

(52) *External topicalization crossing finite CPs and nonfinite TPs*

- a. [Zhe pian baogao], wo xiangxin [_{CP} Lisi xie-wan-le t].
 this CLF report 1SG believe Lisi write-finish-PFV
 ‘This report, I believe that Lisi has already written.’
- b. [Zhe pian baogao], Lisi hui shefa [_{TP} (shuo) zai zhe zhou nei xie-wan t].
 this CLF report Lisi will try COMP at this week in write-finish
 ‘This report, Lisi will try to finish writing this week.’

Chen (2023, to appear) offers compelling arguments (e.g., reconstruction for binding and weak crossover) that external topicalization is derived by pure [*A'*]-probes, whereas internal topicalization is derived by composite [*A/A'*]-probes (cf. van Urk 2015, Lohninger, Kovač & Wurmbrand 2022, Lohninger 2024, Chen 2025). Adopting Chen’s proposal that *C*⁰ in Chinese only hosts [*A'*], the contrast between (51) and (52) can be reduced to the size difference of complement clauses. As illustrated in (53), CPs in type I clauses are phases and enforce successive-cyclic movement via SpecCP, rendering an improper *A/A'-after-A'* chain for internal topicalization. In type II clauses, such CPs are absent, and no improper movement chain is induced. External topicalization, on the other hand, forms a proper *A'-after-A'* chain via CPs and thus may cross type I clauses.

(53) *Internal topicalization and improper movement*

- a. * [_{TP_{matrix}} ... [_{Top_{internal}} DP_{*i*} [_{TOP}, [φ] Top_{internal} [_{TOP+φ}] ... [_{vP} V_{matrix} [_{CP_{embedded}} t_{*i*} C_[_{TOP}] [_{TP} ... V t_{*i*} ...]]
- b. [_{TP_{matrix}} ... [_{Top_{internal}} DP_{*i*} [_{TOP}, [φ] Top_{internal} [_{TOP+φ}] ... [_{vP} V_{matrix} [_{TP_{embedded}} ... V t_{*i*} ...]]

Together, there are three locality profiles of complement clauses. Finite CPs form the most opaque domain for syntactic operations, less so for nonfinite TPs, and even less so for restructuring vPs, which are the most transparent. Their distinction is confirmed by the contrast between three types of cross-clausal dependencies: the most liberal external topicalization, the more restricted internal topicalization, and the most restricted aspect lowering. Such an implicational hierarchy supports the gradient structural distinction of complement

size: vP, TP, and CP (N. Huang 2018, C.-T. J. Huang 2022, He 2024), among which vP stands out in lacking an AspP.²¹

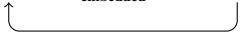
Before proceeding to how the exceptional scopal behavior of *you* forms arguments for the current view, we address the status of semicomplementizer *shuo* below, which has been controversial as pointed out by two anonymous reviewers. While the majority of the literature analyzes it as (phasal or nonphasal) C⁰ (N. Huang 2018, J. Huang 2021; cf. S. Huang 2003, Yeung 2006, Chappell 2008), different views have been raised: see Paul 2014: fn. 27 for problematic cases, Ye 2024 for a compound view for communicative verbs, and Yuan & Saito 2021 for matrix sentence-initial *shuo* as an evidential head. N. Huang (2018: 359), prompted by an anonymous reviewer of his, also suggests that *shuo* can be “a functional head that originates in a lower domain, e.g. the inflectional domain.” He nevertheless also raises several challenges to this alternative (pp. 369–370), along with extensive arguments against other possible categories of *shuo* (pp. 372–374).

If we accept *shuo*’s C⁰ status, however, it would be puzzling why it does *not* block internal topicalization across nonfinite clauses, as we saw above in (51). This is unexpected since CP should host only [A’]-features which renders the composite [A/A’]-movement of internal topicalization improper. This puzzle is also acknowledged by N. Huang (2018: fn. 5) and C.-T. J. Huang (2022: 52). Note that one cannot say that *shuo* only projects nonphasal CPs or [A/A’]-probes; otherwise, internal topicalization across finite clauses with *shuo* would be wrongly predicted to be possible, contradicting (54).

- (54) *Wo [zhe pian baogao] xiangxin [_{CP} shuo Lisi xie-wan-le t].
 1SG this CLF report believe COMP Lisi write-finish-PFV
 Intended: ‘I believe that Lisi has already written this report.’

To resolve this issue: either one adopts an ambiguity view (cf. N. Huang 2018: pp. 367–368), where *shuo*_{finite} is always phasal with [A’] in finite clauses, and *shuo*_{nonfinite} is always nonphasal and/or with [A/A’] in nonfinite clauses; or one follows C.-T. J. Huang’s (2022) suggestion that CPs are “Exfoliated” (in the sense of Pesetsky 2021, 2024) in nonfinite clauses along with their phasehood.²²

We adopt the latter solution and assume the following: (i) Exfoliation may be triggered by an A-dependency (e.g., raising or control) to remove phasal layers that induce a locality issue on that dependency (following Pesetsky 2021, 2024); (ii) Exfoliation can only remove structures without phonological exponence (after C.-T. J. Huang 2022: 54–56). N. Huang (2018: 370–371) and C.-T. J. Huang (2022) convincingly show that *shuo* cliticizes to the preceding matrix verb. There can be no intervening materials in between, as schematized below:

- (55) ...V_{matrix}-*shuo* [_{CP_{embedded}} (*XP) t ...]_{TP} V ...
- 

The CP layer is evacuated and may, though not necessarily, be removed. Since nonfinite clause-taking predicates are either control or raising predicates, in order to establish such dependencies with the embedded PRO/subject across phasal [A’]-CPs, Exfoliation kicks in to remove the CP with its phasehood. As a result,

²¹ Following the cartographic spirit, we do not deny the possibility of more complement clause sizes such as AspP, FinP, ForceP, etc. What we propose is that there is a trichotomy of locality profiles regarding cross-clausal phenomena, and other possible sizes fall into one of the three types (e.g., AspP patterning with TP).

²² Of course, this issue will not arise if one does not accept *shuo*’s C⁰ status and treats it as a functional head in the TP level (e.g., Evid⁰ (cf. Yuan & Saito 2021) or Mood⁰).

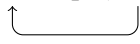
internal topicalization is allowed even with the presence of *shuo*. Exfoliation does not occur in finite clauses since there is no attempted A-dependency. Therefore, internal topicalization is still blocked as in (54).²³

Not only does adopting Exfoliation resolve the puzzle that *shuo* does not alter locality profiles, but it also has a welcome result: nonfinite clauses are necessarily (at most) TPs for cross-clausal A-dependencies. This fits squarely with the size-based view of finiteness.²⁴ In the next section, we return to ‘again’ skipping and argue that it constitutes further support to the current view.

4 ‘Again’ skipping under the size-based approach

Our proposal of ‘again’ skipping consists of two components. The first is that nonfinite clauses can range from as small as vP, which lacks (outer) AspP to as big as TPs, as laid out above. The second is that there are two surface positions for ‘again’ in Mandarin (cf. Lin & Liu 2009): high ‘again’ (*you*) associates with and moves to AspP, as in (56a), whereas low ‘again’ (*zai*) does not move and remains within vP as in (56b).²⁵ Anticipating a split aspect approach (e.g., Tsai 2008, Travis 2010) discussed in Section 6, we clarify that the aspect discussed in this section is the outer aspect.

- (56) a. *High ‘again’*

$$[\text{AspP } you \text{ Asp } [\text{vP } t \text{ V } [\text{VP } \dots]]$$

 b. *Low ‘again’*

$$[\text{AspP } \text{Asp } [\text{vP } zai \text{ V } [\text{VP } \dots]]$$

4.1 Deriving *you*’s exceptional scopal behavior

We suggest that both *you* and *zai* can be base-generated within the vP/VP level as adjuncts, but they surface in different positions (cf. Lin & Liu 2009). *You* surfaces at the AspP level as a result of its association with AspP by movement (following T. Xu 2012, 2016 and Liu 2021, 2024).²⁶ *Zai*, on the other hand, stays within the vP and

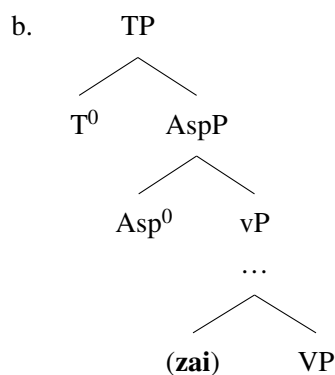
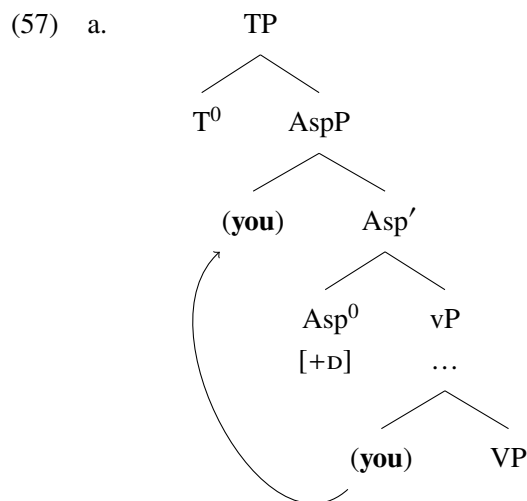
²³ A prediction is that when a raising predicate takes a finite CP, hyperraising should be possible after Exfoliation. This is precisely the claim made by Pesetsky (2021: §3.4): some languages like Lusaamia (Bantu) allows raising from an embedded clause with finite morphology but not with complementizers. Interestingly, hyperraising is attested in Chinese as argued by Chen (2023, 2025) for Mandarin and by Lee & Yip (2024a) for Cantonese, although their proposal does not involve Exfoliation. We leave this issue for future research.

²⁴ If one adopts the ambiguity view of *shuo*, the current size-based approach would need to be weakened so that nonfinite CPs are possible (i.e., neutralizing type I and type II clauses). However, our main claim that nonfinite clauses must be differentiated by size still holds (i.e., type II vs. III clauses). Note that under the ambiguity view, the occurrence of SAY-complementizers in finite and nonfinite clauses across Chinese languages (e.g., Cantonese *waa* in Yeung 2006, J. Huang 2021 and Teochew *tan* in Huang & Cai 2025, both of which have different lexical sources from *shuo*) would be attributed to the lexical ambiguity *accidentally* shared by their lexicons and moreover need to cluster with (non)phasehood and [A’]- vs. [A/A’]-features in the same way.

²⁵ We assume that head movement of V^0 to v^0 takes place postsyntactically at the PF. See Chomsky 2000 for more detailed discussion and Cheng, Yi & Xiong 2016 for a related proposal with PF lowering in Chinese. This theoretical choice does not bear on our proposal, though. For expository purposes, we do not show PF movement throughout the paper.

²⁶ Some may question the mobility of the adverb *you*. As defended by T. T.-M. Lee (2024b), the Cantonese counterpart *jau* can move across a quantificational subject to gain semantic effects, whose pattern can be reduplicated with Mandarin *you*, as shown below with an exclusive focus subject:

does not associate with the AspP above.²⁷ The positional difference is evidenced by their (in)compatibility with aspectual elements in Section 2.2 (correlation II in (36)). Concretely, we follow Lin & Liu’s (2009) proposal that *you* is licensed by [+D] on the closest Asp⁰, formulated here as feature checking with an unvalued dynamic feature [*u*+D] on *you*. This feature is absent on *zai*, as we have seen from the embedding patterns presented in Section 2.2 (correlation I in (30)). The structural difference between *you* and *zai* is diagrammed in (57).



As discussed in Section 2.2, one key motivation of [+D] licensing of *you* in Lin & Liu 2009 comes from *you*’s resistance to embedding under predicates like *xiang* ‘want’, which cannot embed perfective *-le* with [+D] either. This licensing should be syntactic rather than semantic in nature, since *you* can “skip” *xiang* ‘want’ to

-
- (i) a. Jiu [yi ge ren]_F **you** kaodao-le hao chengji.
 just one CLF person again test-PFV good score
 ‘Just one person again got a good score. (Everyone else did not get a good score this time, although they may have got a good score last time.)’
- b. **You** jiu [yi ge ren]_F kaodao-le hao chengji.
 again just one CLF person test-PFV good score
 ‘Again, just one person got a good score. (Everyone else did not get a good score last time and this time).’

See T. T.-M. Lee 2024b for detailed arguments of the movement analysis.

²⁷ *Zai* can also move (from low projections like VPs), although to a vP-internal position, which is identified as an inner aspect projection in Section 6. What suffices here is that *zai* does not move out of vP.

only take scope over the embedded $[-D]$ predicate. In other words, *you* can in principle be interpreted with a $[-D]$ predicate, as long as *you* itself is syntactically licensed by some $[+D]$ Asp⁰. We take Lin & Liu's (2009) $[+D]$ licensing at face value and will demonstrate how this component makes the right prediction on our data below.²⁸

Let us see how the proposal derives the narrow scope readings of *you*, such as the one in (58) (reproduced from (12a)).²⁹

- (58) Xiaoming **you** xiang [qu Xianggang].
 Xiaoming again want go Hong.Kong
 'Xiaoming wants to again go to Hong Kong.'

The derivation of (58) is sketched in (59).³⁰ First, the embedded clause is vP-sized and does not contain an AspP. Second, *you* moves to the specifier position of the *matrix* AspP in overt syntax in order to value its dynamic feature. Third, at LF, *you* reconstructs to the base-generated position (indicated by the frameboxed trace) to yield the narrow scope reading, "skipping" the scope of the matrix predicate 'want'. This is how the exceptional scopal behavior across restructuring clauses is derived.³¹

²⁸ A reviewer raises a question on the origin of $[+D]$. We speculate that this has to do with the diachronic development of 'again' morphemes. Apparently, the repetitive use of *you* arises as early as in the Spring and Autumn and the Warring States periods (c. 770–221 BCE), as in (1a), whereas that of *zai* starts quite recently, stemming from its original usage as indicating a frequency 'twice', as in (1b).

- (i) a. Qiu, Guo ren qin Jin. Dong, Guo ren **you** qin Jin. Old Chinese
 autumn Guo people invade Jin winter Guo people again invade Jin
 'In autumn, the people of Guo invaded Jin. In winter, the people of Guo invaded Jin again.'
 (*Zuo zhuan* • *Zhuang Gong er-shi-liu nian* [*Commentary of Zuo* • 26th year of Duke Zhuang] c. before 221 BCE)
- b. Yi gu zuo qi, **zai** er shuai, san er jie. Old Chinese
 first drum stir.up spirit second and wane third and exhaust
 'The first round of drum stirred up the spirits, which waned in the second and exhausted in the third.'
 (*Zuo zhuan* • *Zhuang Gong shi nian* [*Commentary of Zuo* • 10th year of Duke Zhuang] c. before 221 BCE)

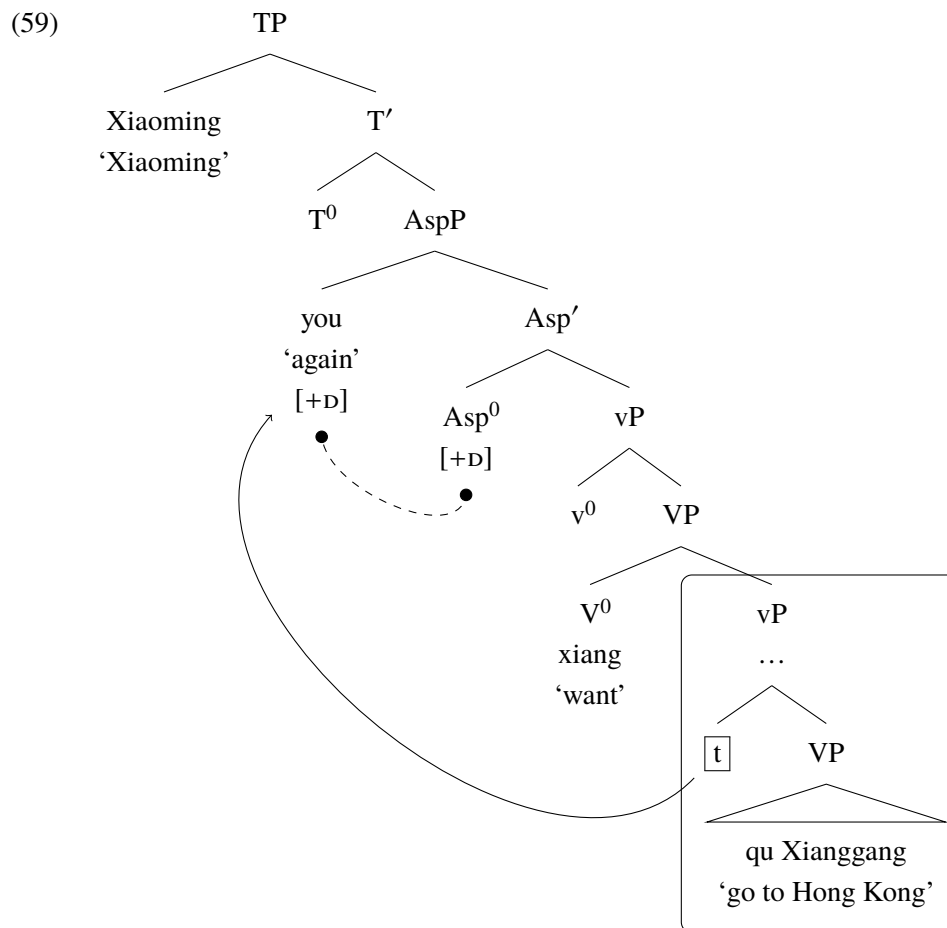
Given the difference in when their repetitive use emerges, we speculate that *you*'s sensitivity to the dynamicity feature is a diachronic result of the competition between *you* and *zai* after the repetitive use of *zai* arises. This idea is based on the observation that in Old Chinese, when *zai* has not yet developed its repetitive use, *you* does not appear to be sensitive to $[\pm D]$. For instance, below are cited fragments where *you* occurs in the complement of modals, which is $[-D]$ rather than $[+D]$:

- (ii) a. Bi xi, bi **you** qing di yu zhuhou.
 3sg happy must again ask.for land from feudal.lord
 '[If] he is happy, he will certainly ask for land again from the feudal lords.'
 (*Shuo yuan* • *Quanmou* [*Garden of stories* • *Trickery*] c. before 6 BCE)
- b. Sizhe jiang qu, bu ke **you** de.
 dead will go not can again find
 'The dead will go and cannot be found again.'
 (*Baihu tongyi* • *Beng hong* [*General meaning of White Tiger Hall discussions* • *Deaths of emperors and feudal lords*] c. 57–88 CE)

²⁹ The movement and reconstruction analysis of *you* raises a learnability question as to how children acquire the ambiguity of *you* when all its possible readings surface in the same surface position. We refer interested readers to Xu, Snyder & Christie 2022, 2024 for a comprehensive discussion on this.

³⁰ Head movement and subject movement are omitted for simplicity.

³¹ An astute reviewer has noticed a prediction by the movement and reconstruction analysis of *you* with respect to its interaction with a quantifier between the matrix and embedded aspect heads — namely, with *you* surfacing in the matrix and a quantifier occupying a



In comparison, when the matrix verb is a finite clause-taking predicate like ‘believe’, the embedded CP must have a [+D] AspP, capable of licensing *you*. Since *you* cannot move to the matrix AspP, high pronunciation is always a result of base generation in the *matrix* clause. There would be no trace in the embedded clause to reconstruct back to, hence the lack of narrow scope across CP boundaries. This explains the scopal contrast of *you* across finite vs. nonfinite clauses (i.e., generalization II (20) in [Section 2.1](#)).

position between the two aspect heads (e.g., the embedded subject position), one could get a reading in which the matrix verb scopes over the quantifier, which further scopes over ‘again’. This prediction is borne out:

- (i) (You’re an insurance salesperson, and whenever you meet someone new, the first thing you do is try to sell them insurance. Your friend, who knows you very well, is introducing you today to a group of people you’ve never met before. They’ve already bought the insurance you’re selling, but not because of you, since you didn’t even know them until now. However, knowing you’re likely to start pitching your insurance again, unprompted, your friend says this to you.)

Tamen suoyou ren dou mai-guo zhe ge baoxian le. Ni bu yao you quan suoyou ren mai!
 3PL all person all buy-EXP this CLF insurance PRF 2SG not IMP again urge all person buy
 ‘All of them have bought this insurance. Don’t urge everyone to buy again!’

- (60) *Low interpretation of you with a finite clause-taking predicate*
 (Intended:) ‘(Xiaoming) believes that there again appeared coronavirus.’ (cf. (17))

a. *Low pronunciation: OK*

[AspP_{matrix} ... [vP ‘believe’ [CP ... [AspP_{embedded} *you* Asp_[+D] [vP t ‘coronavirus appeared’]]]]]]

b. *High pronunciation: **

*[AspP_{matrix} *you* ... [vP ‘believe’ [CP ... [AspP_{embedded} Asp_[+D] [vP t ‘coronavirus appeared’]]]]]]

Unlike *you*, *zai* does not move to AspP above vP, and there is no reconstruction of *zai* across restructuring clause boundaries. It is always interpreted locally in its base-generated position at the vP level and does not exhibit exceptional scopal behavior. In other words, it displays a “what you see is what you get” scope, as in (61). Up to this point, we have derived generalization I (16) in Section 2.1.

- (61) a. *High pronunciation: High interpretation*

Xiaoming hui [AspP [vP_{matrix} *zai* [vP xiang [vP_{embedded} qu Xianggang]]]].
 Xiaoming will again want go Hong.Kong
 ‘Xiaoming will again want to go to Hong Kong.’

b. *Low pronunciation: Low interpretation*

Xiaoming [AspP [vP_{matrix} xiang [vP_{embedded} *zai* [vP qu Xianggang]]]].
 Xiaoming want again go Hong.Kong
 ‘Xiaoming wants to again go to Hong Kong.’

4.2 Predictions for nonfinite clauses larger than vP

Recall from Section 3.1 that (i) the semicomplementizer *shuo*, (ii) the adverb *ye* ‘also’, and (iii) internal focus fronting signal a complement that is as large as a TP and thus contains AspP. Since AspP triggers (defective) intervention effects to aspect lowering, we expect it to also intervene the movement of *you* to matrix AspP. We therefore predict that these TP-signaling elements should block the exceptional scopal behavior of *you*. Our prediction is borne out, as illustrated below.

In (62), we have the context from (12a)/(58), where Xiaoming wants to go to Hong Kong now, went to Hong Kong before, but did not want to go to Hong Kong before. The example here is minimally different from the baseline example (12a)/(58) in that its embedded clause is now headed by the semicomplementizer *shuo*. Here, while the wide scope reading of *you* remains infelicitous (as it is not supported by the context), the narrow scope reading becomes unavailable.³²

³² C.-T. J. Huang (2022) judges examples along the lines of (62) to be acceptable, but our consultants reported uniform unacceptability. This tension also applies for the upcoming examples (64–65), which concern the adverb *ye* ‘also’ and internal foci. It is possible that there exist dialect variations between Mainland Mandarin and Taiwan Mandarin, which we leave for future research.

(62) *Semicomplementizer shuo blocking you skipping*

(Yesterday, Xiaoming took a business trip to Hong Kong for work, despite not wanting to travel, as he was compelled by his boss. Unfortunately, he did not complete the work before returning. Today, worried about getting fired for not finishing the work, he wants to go back to Hong Kong again to finish it.)

#Xiaoming **you** xiang [shuo qu Xianggang].

Xiaoming again want COMP go Hong.Kong

*Narrow scope: ‘Xiaoming wants to again go to Hong Kong.’

#Wide scope: ‘Xiaoming again wants to go to Hong Kong.’

We attribute this unavailability of the narrow scope reading to the clause size of the embedded clause: as in (63), when the semicomplementizer *shuo* is present, the embedded clause must be (as least) TP (cf. Section 3.2) and include an AspP. The embedded Asp⁰ only carries [-D] and cannot license *you*, but it still induces defective intervention effects and blocks the movement to matrix AspP (as well as agreement with Asp⁰).

(63) *Defective intervention effects to you movement*

*[AspP_{matrix} **you** Asp_[+D] ... [vP ‘want’ [TP *shuo* ... [AspP_{embedded} Asp_[-D] [vP t ‘go to HK’]]]]]

↑ x ↓

As for testing whether the adverb *ye* ‘also’ blocks *you* skipping, the context of interest should involve at least two parallel events of someone having gone to Hong Kong reluctantly. A context like this is provided in (64), where the example is minimally different from (12a)/(58) in that its embedded clause now contains *ye*.³³

(64) *Adverb ye ‘also’ blocking you skipping*

(The day before yesterday, Xiaoming and Xiaohong’s boss forced Xiaohong to go to Hong Kong, and she came back. Yesterday, the boss forced Xiaoming to go to Hong Kong, and he came back. Today, the boss forced Xiaohong but not Xiaoming to go to Hong Kong. However, secretly admiring Xiaohong, Xiaoming wants to go to Hong Kong again today.)

#Xiaoming **you** xiang [ye qu Xianggang].

Xiaoming again want also go Hong.Kong

*Narrow scope: ‘Xiaoming wants to again go to Hong Kong, too.’

#Wide scope: ‘Xiaoming again wants to go to Hong Kong, too.’

Once again, we observe that the narrow scope reading becomes unavailable, aligning with our predictions. A similar observation is made with internal foci:

³³ When *you* ‘again’ is absent, (64) with *ye* ‘also’ is felicitous under the given context. The same goes for (65) with internal focus.

(65) *Internal focus blocking you skipping*

(In his undergraduate years, Xiaoming was a very lazy student and was bad at math. He hated taking math courses, but he was forced by his advisor to take calculus. When Xiaoming moved on to pursue a PhD, however, he became an overachieving student and developed an obsession with math. Now, he does want to take calculus.)

#Xiaoming **you** xiang [lian weijifen dou xuan].

Xiaoming again want even calculus all choose

*Narrow scope: ‘Xiaoming wants to again choose even calculus.’

#Wide scope: ‘Xiaoming again wants to choose even calculus.’

We summarize these phenomena and their (in)compatibility with aspect lowering and *you* skipping in Table 3.

Table 3 Finiteness tests, their clause sizes, and whether they allow aspect lowering and *you* skipping in Mandarin

	Clause size	Aspect lowering	<i>You</i> skipping
Clause selected by ‘believe’/‘say’	CP	✗	✗
Semicomplementizer <i>shuo</i>	TP	✗	✗
Adverb <i>ye</i> ‘also’	TP	✗	✗
Internal focus	TP	✗	✗
None of the above	vP	✓	✓

4.3 Further specifications of predicate selection

In closing this section, we fully spell out the additional specifications that we make on the selectional constraints of different types of embedding predicates. As discussed in Section 3, we follow previous works and claim that finite clauses are CPs, while nonfinite clauses are vPs or TP. We further specify that when a nonfinite clause-taking predicate selects a complement larger than AspP, they distinguish whether their semantics tolerates a dynamic ([+D]) aspect feature.

Specifically, *xiang* ‘want’ is semantically incompatible with [+D] in its complement. It selects a nonfinite clause that is either a vP or a TP with a [−D] AspP. In contrast, *rang* ‘let’ is flexible and allows for its complement to contain vP or [±D] AspPs. As discussed in Section 2.2, *xiang* and *rang* differ in whether they can embed the perfective suffix *-le*. On the other hand, *xiangxin* ‘believe’ represents a finite clause-taking predicate that must take a CP, which contains a TP and AspP with [+D] or other aspectual features like [EXP]. We summarize their selectional constraints in Table 4.

We have seen in Section 2.2 that the complement clause of *rang* ‘let’ can take [+D] AspP: it may embed *you*. An example is below in (66). Here, *you* must agree with the closest [+D] AspP⁰.

- (66) a. Zhangsan rang Lisi [**you** qu yi ci Xianggang].
 Zhangsan let Lisi again go one time Hong.Kong
 ‘Zhangsan let Lisi again go to Hong Kong once.’

Table 4 Selectional constraints of different predicates in Mandarin

	vP	TP with [-D] AspP	TP with [+D] AspP	CP
<i>Xiang</i> ‘want’	✓	✓	#	*
<i>Rang</i> ‘let’	✓	✓	✓	*
<i>Xiangxin</i> ‘believe’	*	*	*	✓

- b. [_{AspP_{matrix}} ... [_{vP} ‘let Lisi’ [_{TP} ... [_{AspP_{embedded}} *you* Asp_[+D] [_{vP} t ‘go to HK’]]]]]
-

We then predict that aspect lowering should be unavailable in sentences like (66), since the presence of [+D] Asp⁰, while not enough to license *-guo* because of the lack of [EXP], still blocks agreement between *-guo* and the matrix Asp⁰ due to defective intervention.

This prediction is correct, as shown in (67). Here, the context supports a reading where (i) *you* takes narrow scope, whereas (ii) the experiential aspect takes wide scope. That is, (i) the presupposed antecedent event is with respect to the embedded predicate (i.e., Lisi went to Hong Kong before), and (ii) the matrix predicate is realized so that there is no entailment of the embedded predicate’s actuality (i.e., Lisi did not *again* go to Hong Kong). With matrix *you* and embedded *-guo* in (67a), the intended reading is obtained. In other words, both aspect lowering and *you* skipping are possible simultaneously. When *you* is embedded in (67b), however, aspect lowering becomes unavailable.

(67) *Wide scope context: Exp > ‘let’ > ‘again’ > ‘go’*

(Zhangsan is Lisi’s boss. Two years ago, Lisi went to Hong Kong for traveling. It was not Zhangsan who let him do that; Lisi himself wanted to go there. Last year, Zhangsan asked Lisi to go to Hong Kong, but Lisi declined and didn’t make the trip.)

a. *Aspect lowering and matrix you: OK*

Zhangsan **you** rang Lisi [qu-guo yi ci Xianggang].

Zhangsan again let Lisi go-EXP one time Hong.Kong

Wide scope of *-guo* and narrow scope of *you*: ‘Zhangsan has let Lisi again go to Hong Kong once.’

b. *Aspect lowering and embedded you: #*

#Zhangsan (cengjing) rang Lisi [**you** qu-guo yi ci Xianggang].

Zhangsan before let Lisi again go-EXP one time Hong.Kong

#Narrow scopes of *you* and *-guo*:

‘Zhangsan lets Lisi have again gone to Hong Kong once (before).’

*Wide scope of *-guo* and narrow scope of *you*:

‘Zhangsan has let Lisi again go to Hong Kong once (before).’

The above contrast falls out from the current analysis. Both aspect lowering and *you* skipping are possible in (67a), because the embedded clause size is a vP: there is no embedded AspP to intervene. On the contrary, there is an embedded [+D] AspP to which *you* moves locally in (67b). This [+D] AspP lacks [EXP] and acts as a defective intervener to *-guo*’s agreement with the matrix AspP. The diagrams in (68a) and (68b) illustrate (67a) and (67b), respectively.

- (68) a. $[\text{AspP}_{\text{matrix}} \text{you Asp}_{[+D, \text{EXP}]} [\text{vP}_{\text{matrix}} \text{'let Lisi' } [\text{vP}_{\text{embedded}} \text{t 'go' -guo ...}]]]$
- b. $*[\text{AspP}_{\text{matrix}} \text{Asp}_{[\text{EXP}]} [\text{vP}_{\text{matrix}} \text{'let Lisi' } [\text{TP}_{\text{embedded}} [\text{AspP}_{\text{embedded}} \text{you Asp}_{[+D]} [\text{vP}_{\text{embedded}} \text{t 'go' -guo ...}]]]]]$

5 Postverbal ‘again’ lowering in Cantonese

Turning to Cantonese, apart from the two preverbal ‘again’ adverbs *jau* (counterpart of Mandarin *you*) and *zoi* (counterpart of Mandarin *zai*), it also has two postverbal ‘again’-type elements: verbal suffixes *-faan* and *-gwo*.³⁴ We show that the two elements exhibit similar asymmetries in exceptional scopal behavior, which is constrained by the two syntactic factors that we have identified for Mandarin: (i) the size of embedded clauses and (ii) the position of ‘again’. This strikingly parallel pattern supports a general cartography of ‘again’ elements under the split aspect approach.

To begin with, Cantonese verbal suffix *-faan* presumes that a related event has occurred before. It has a repetitive reading as well as resumption and restitutive readings (Zhan 1958, Cheung 1972, 2007, Chang 1996, Peng 1999, 2010, S.-W. Tang 2001, Fong 2007), as illustrated in (69). We focus on the repetitive reading in this paper.³⁵

(69) Verbal suffix *-faan* in Cantonese

- a. Ngodei hai-**faan** gosyu ginmin. repetitive
 1sg at-again there meet
 ‘Let’s meet there again.’
 (Peng 2010: 96)

³⁴ Unlike Mandarin *you*, Cantonese *jau* can occur in nonfinite clauses embedded under ‘want’ (i.e., with a $[-D]$ Asp^0):

- (i) Aaming soeng [**jau** heoi do ci Hoenggong].
 Ming want again go more time Hong.Kong
 ‘Ming wants to go to Hong Kong one more time.’

As far as we know, Cantonese *jau* does not have the ‘again’ skipping behavior:

- (ii) Aaming **jau** soeng [heoi Hoenggong].
 Ming again want go Hong.Kong
 *Narrow scope: ‘Ming wants to again go to Hong Kong.’ *‘want’ > ‘again’ > ‘go’
 Wide scope: ‘Ming again wants to go to Hong Kong.’ ‘again’ > ‘want’ > ‘go’

This follows from our proposal. The embeddability of *jau* under ‘want’ shows that it does not carry $[u+D]$. Thus, *jau* does not move to an (outer) aspect position from a nonfinite clause, and can only be base-generated in the matrix clause when it is pronounced there. Hence, no reconstruction to the embedded clause can occur (i.e., no “skipping” of the matrix verb). The variations between Cantonese and Mandarin are reduced to a difference in their lexicons: only Mandarin *you* is specified with $[u+D]$, but not Cantonese *jau*.

³⁵ We do not discuss two other homophonous *-faan* morphemes in this paper. The first one has a modal favor “attaining ideal state” (cf. S.-W. Tang 2001). It does not presuppose a past event and is not an ‘again’-type element. The second one is a directional complement (i.e., not a verbal suffix) as in *fei faan* ‘fly back’ (similar to *hui* ‘return’ in *fei hui* ‘fly back’ in Mandarin, cf. footnote 41). While this *-faan* is historically related to the ‘again’ suffix *-faan* (e.g., Peng 2010), they are distinct synchronically: only the directional complement *-faan* can be suffixed by perfective *-zo* and infixes by *-dak* ‘can’ or *-m* ‘not’.

- b. Nei zou-**faan** nei ge je, ngo tai-**faan** ngo ge syu. *resumption*
 2SG do-again 2SG GE thing 1SG read-again 1SG GE book
 ‘You (continue) do(ing) your thing; I (continue) read(ing) my book.’
 (Zhan 1958: 121)
- c. Zap-**faan** zi bat bei ngo. *restitutive/counterdirectional*
 pick.up-again CLF pen to 1SG
 ‘Pick up the pen (that fell on the floor) for me.’
 (Zhan 1958: 121, adapted)

The other suffix *-gwo* also presumes that a related event has occurred but specifies a different relation between the previous and present events. *-Gwo* in (70) expresses a repetitive reading with a sense of “fixing” undesirable outcomes of previous events (Cheung 1972, 2007, Chang 1996, Chen & Lin 2006, Yan 2009). This use should be distinguished from the experiential aspect use of *-gwo*.³⁶

- (70) *Verbal suffix -gwo in Cantonese*
 Mgoi nei se-**gwo** keoi laa1! *repetitive*
 please 2SG write-again 3SG SFP
 ‘Please rewrite it (to get it right)!’
 (Cheung 2007: 156)

As we will see below, only *-faan* but not *-gwo* allows wide scope across a (nonfinite) clause boundary over the matrix verb, as schematized in (71). That is, ‘again’ is pronounced low but interpreted high. We refer to this phenomenon as ‘*again*’ lowering (or *-faan* lowering), a mirror image of the preverbal *you* in Mandarin, which may be pronounced high and interpreted low (“skipping”).

- (71) *Cantonese postverbal ‘again’ lowering*
- | | | |
|---------------------|--|-------------------|
| HIGH INTERPRETATION | $\dots [V_{\text{matrix}} [V_{\text{embedded}} \text{‘again’} \dots]$ <div style="text-align: center; margin-top: -10px;"> $\uparrow$ </div> | LOW PRONUNCIATION |
|---------------------|--|-------------------|

The relevant examples are given in (72a), where both suffixes are embedded in a nonfinite clause. The embedded verb *deoilam* has a resultative complement *lam* ‘die, (lit.) fall’ and denotes an event that cannot be repeated (i.e., ‘killing (the boss)’). While *-faan* may take scope over the matrix verb ‘want’ and give the plausible wide scope reading of repeated desire in (72a), *-gwo* cannot, and only gives the infelicitous narrow scope reading ‘killing the boss again’ in (72b).³⁷

- (72) *Embedded ‘again’ taking wide scope over ‘want’ in Cantonese*
 (When Ming was a gangster, he always wanted to murder his maniac boss, though he never tried to.

³⁶ It is common in Chinese languages that an experiential aspectual suffix is homophonous with a repetitive suffix, such as in Hakka, Hokkien/Southern Min, Gan, Southwestern Mandarin, Wu, and Xiang, among others (Wen 2020). This suggests some potential grammaticalization path from experiential aspect to the repetitive use, which is beyond the scope of this paper.

³⁷ The scopal interaction is not limited to matrix predicates. In the presence of other quantificational elements like the universal suffix *-saai* (e.g., P. P.-I. Lee 2012), *-faan*’s exceptional scope is wider than *-saai*’s scope in biclausal configurations, though *-saai* is further away from the verbal root and outscopes *-faan* in monoclausal configurations. We thank an anonymous reviewer for raising this point.

- (i) Aaming (jau) soeng [deoilam-**faan-saai** keoi di leng].
 Ming again want kill-again-all 3SG CLF.PL sidekick
 ‘Ming again wants to kill all of his sidekicks.’ *again > want > all*

He no longer wanted so after he left the gang. Today, he met his boss on the street, who insulted and slapped him. Now, Ming is so angry that he wants to kill him again.)

- a. Aaming soeng [deoilam-**faan** keoi daailou].
 Ming want kill-again 3sg boss
 #Narrow scope: ‘Ming wants to again kill his boss.’ #want > again
 Wide scope: ‘Ming again wants to kill his boss.’ again > want
- b. #Aaming soeng [deoilam-**gwo** keoi daailou].
 Ming want kill-again 3sg boss
 #Narrow scope: ‘Ming wants to again kill his boss.’ #want > again
 *Wide scope: ‘Ming again wants to kill his boss.’ *again > want

Note that *-faan* may still give a narrow scope reading: it is scopally ambiguous. The two readings can be disambiguated by placing the preverbal *jau/zoi* ‘again’ before the respective verbs, as in (73a). In contrast, *-gwo* is not scopally ambiguous and has only narrow scope. With matrix *jau/zoi* in (73b), the only reading is a multiple ‘again’ reading with the matrix verb’s scope sandwiched between the two ‘again’ elements (i.e., *jau/zoi* > want > *-gwo*).³⁸

(73) [Context: same as (72)]

- a. Aaming {i. *jau/zoi*} soeng [{ii. #*jau/#zoi*} deoilam-**faan** keoi daailou].
 Ming again want again kill-again 3sg boss
 #Narrow scope in (ii): ‘M. wants to again kill his boss.’ (#want>again)
 Wide scope in (i): ‘Ming again wants to kill his boss.’ (again>want)
- b. #Aaming {i. *jau/zoi*} soeng [{ii. *jau/zoi*} deoilam-**gwo** keoi daailou].
 Ming again want again kill-again 3sg boss
 #Narrow scope in (ii): ‘M. wants to again kill his boss.’ (#want>again)
 #Sandwiched scope in (i): ‘Ming again wants to again kill his boss.’ (#again>want>again)

Note further that the matrix predicate *soeng* ‘want’ cannot be suffixed by *-faan* or *-gwo*, as illustrated in (74). One may wonder whether such exceptional scopal behavior is a morphological repair for predicates that cannot take verbal suffixes.

- (74) Aaming soeng(***-faan**/***-gwo**) deoilam keoi daailou.
 Ming want-again kill 3sg boss
 Int.: ‘Ming again wants to kill his boss.’

For predicates that may take verbal suffixes such as *hyun* ‘urge’ in (75), however, *-faan* lowering is still possible, as shown by *-faan*’s wide scope readings over ‘urge’ in (76a). *-Gwo*, again, lacks such wide scope readings, as in (76b). Thus, the scopal effects are not the result of some last-resort type morphological processes.

³⁸ The narrow and sandwiched scope readings can be made felicitous in a game world where the characters may resurrect after being killed.

(75) (I used to urge Ming to apply for a PhD. Yet, he kept ignoring me and even became mad at me, so I just gave up. Today, our teacher and I see that the government has additional funding for PhD studies, ...)

- a. Soji ngo (jau) hyun-faan Aaming [bou PhD].
 so 1sg again urge-again Ming apply PhD
 Wide scope: 'So I again urge Ming to apply for PhD.' (again>urge)
- b. Soji lousi giu ngo (zoi) hyun-gwo Aaming [bou PhD].
 so teacher ask 1sg again urge-again Ming apply PhD
 Wide scope: 'So our teacher asks me to again urge Ming to apply for PhD.' (again>urge)

(76) Embedded 'again' takes wide scope over 'urge' in Cantonese
 (Same context with (75))

- a. Soji ngo (jau) hyun Aaming [bou-faan PhD].
 so 1sg again urge Ming apply-again PhD
 #Narrow scope: 'So I urge M. to reapply for PhD.' (#urge>again)
 Wide scope: 'So I again urge Ming to apply for PhD.' (again>urge)
- b. #Soji lousi giu ngo hyun Aaming [bou-gwo PhD].
 so teacher ask 1sg urge Ming apply-again PhD
 #Narrow scope: 'So our teacher asks me to urge Ming to reapply for PhD.' (#urge>again)
 *Wide scope: 'So our teacher asks me to again urge Ming to apply for PhD.' (*again>urge)

We have seen cases where *-faan* takes wide scope across a nonfinite clause boundary. Its exceptional wide scope may even be available across multiple nonfinite clause boundaries. In (77), the deeply embedded *-faan* scopes over the matrix verb 'want' across the intermediate verb 'ask' and the embedded verb 'kill'.

(77) Embedded 'again' takes wide scope across multiple nonfinite clause boundaries in Cantonese
 (Ming is an assassin in a gang and I have a personal grudge against his boss. I have thought multiple times to hire Ming to assassinate his boss, but never done so. Today, I met his boss on the street and he insulted and slapped me. Now, I again want to ask Ming to kill him.)
 Ngo (jau) soeng [giu Aaming [deoilam-faan keoi daailou]].
 1sg again want ask Ming kill-again 3sg boss
 'I again want to ask Ming to kill his boss.' (again>want>ask>kill)

The exceptional scope behavior of postverbal 'again' is constrained by two syntactic factors. First, the size/finiteness of the embedded clauses matters. *-faan* lowering is not possible with finite-clause-taking predicates like *seon* 'believe' in (78a). The wide scope reading can only be obtained by attaching *-faan* to the matrix verb, as in (78b).

(78) Embedded 'again' not taking wide scope over 'believe' in Cantonese
 (Ming quit being a Christian years ago. Today, he had a traffic accident, and heard God's voice when he was badly injured. Now, He once again believes that God exists.)

- a. #Aaming seon [jau-faan san].
 Ming believe have-again God
 #Narrow scope: 'Ming believes that there is again God.' #believe > again
 *Wide scope: 'Ming again believes that there is God.' *again > believe

- b. Aaming seon-faan [jau san].

Ming believe-again have God

Wide scope: 'Ming again believes that there is God.'

again > believe

The sensitivity to finiteness is exactly what we have seen for the scope skipping effect of *you*. Such contrasts hold across a wide range of clause-taking predicates: the wide scope interpretation of *-faan* is only allowed with nonfinite-clause-taking predicates, as listed in (79). Finite-clause-taking predicates disallow such exceptionally wide scope and are listed in (80).

- (79) Nonfinite-clause taking verbs allow embedded -faan's exceptional scope

E.g., *bik* 'force', *hyun* 'persuade/urge', *giu* 'ask', *ceng* 'invite', *paai* 'send', *soeng* 'want', *daasyun* 'intend', *gaiwaak* 'plan', *zeonbei* 'prepare', *hoici* 'start', *hoji* 'may', *gam* 'dare', *hang* 'be willing', ...

- (80) Finite-clause taking verbs disallow embedded -faan's exceptional scope

E.g., *jingwai* 'consider', *soengseon* 'believe', *geidak* 'remember', *gong* 'say', *syunbou* 'declare', *honang* 'be possible', ...

Second, there is also a positional difference between *-faan* and *-gwo* with respect to aspect. Concretely, *-faan* occupies an outer suffix slot in the verbal template in Cantonese, whereas *-gwo* occupies an inner slot. This echos the pattern in Mandarin where "skipping"-*you* is structurally higher than "non-skipping"-*zai*. For instance, *-faan* is incompatible with (outer) aspectual suffixes. In (81), *-faan* can neither follow nor precede the perfective suffix *-zo*.

- (81) *-Faan cannot follow or precede aspectual suffixes*

Ngo jau **tai(*-faan)-zo(*-faan)** ni bun syu.

1SG again read-again-PFV-again this CLF book

Int.: 'I have read this book again.'

In contrast, *-gwo* preceding *-zo* (i.e., *-gwo-zo*) is much more acceptable than *-gwo* following *-zo* (i.e., **-zo-gwo*), as given in (82). (83) gives two naturally occurring examples from the internet for the *-gwo-zo* sequences. We take the difference between (81) and (82)/(83) to indicate that *-faan* appears in a slot that is associated with a higher position (e.g., outer aspect, as will be argued below) and *-gwo* occupies a slot that is associated with a lower position.

- (82) -Gwo may precede aspectual suffixes

Ngo zoi **tai(?-gwo)-zo(*-gwo)** ni bun syu dojatci.

1SG again read-again-PFV-again this CLF book one.more.time

'I have read this book one more time again.'

- (83) a. ... cungsan **se-gwo-zo** Baaijanzung ge daisaam dyun goci.
again write-again-PFV Bayernhymne GE third line lyrics
'rewrote the third line of *Bayernhymne*'s lyrics'

(Wikipedia, 2013-8-26)

- b. Auntie go dinsi waai-zo, **maai-gwo-zo** zek jaagei cyun LG daai Time Machine
 auntie CLF TV break-PFV buy-again-PFV CLF around.20 inch LG with time machine
 lukjing gungnang ge.
 record function GE
 ‘Auntie’s TV is broken. She bought another one, which is around 20 inches and has the Time Machine recording function.’

(Social media, 2018-8-26)

We further observe an asymmetry in the compatibility with *inner* aspect, as instantiated by *phase complements* (PCs). To give some background, PCs refer to postverbal elements expressing the phase/stage of an action (Chao 1965:446–450), such as *hao* in *suo-hao* ‘locked properly’ (*so-hou* in Cantonese), *zhao* in *shui-zhao* ‘slept’ (*fan-zoek* in Cantonese), and *diao* in *mai-diao* ‘sell off/away’ (no Cantonese counterparts), among others (see the Appendix for lists of PCs in Cantonese and Mandarin and tests to distinguish them from resultative verbal complements and suffixes). We refer to them as “inner aspect” since they *precede* outer aspectual suffixes (Tsai 2008, Sybesma 2017, Lu, Lipták & Sybesma 2019), such as perfective *-le/-zo* in (84).³⁹

(84) *PCs preceding aspectual suffixes*

- a. Women mai-**diao-le** fangzi. [Mandarin]
 1PL sell-PC-PFV house
 ‘We finally sold the house.’ (adapted from Sybesma 2017:303)
- b. Keoi fan-**zoek-zo**. [Cantonese]
 3SG sleep-PC-PFV
 ‘S/he has slept.’ (Cheung 2007:114)

Turning to Cantonese *-faan* and *-gwo*, only *-faan* is able to follow PCs, as illustrated in (85) from the internet.⁴⁰ *-Gwo* not only cannot follow PCs, but it also cannot precede PCs, as in (86). In other words, *-gwo* is incompatible with PCs.

³⁹ In Sybesma (2017) and Lu, Lipták & Sybesma (2019)’s terminology, “inner aspect” refers to aspectual projections below vP, including “realization” *-le* in Mandarin (Asp3P, highest), PCs (Asp2P, middle), and resultative complements (Asp1P, lowest). In this paper, we follow Tsai (2008: p. 683) in preserving the term “inner aspect” for PCs. For the position of *-le*, see the discussion in Section 6. The position of resultative verbal complements (RVCs) is not relevant to the current paper, but see Appendix for discussion.

⁴⁰ We notice that *-faan* may also precede PCs, as in (i). Verbal suffixes in Cantonese seem to have some flexibility between suffixation and “infixation” as in (ii), forming the so-called “discontinuous predicates” or “separable verbs”. This holds even for monomorphemic verbs like *feilou* ‘fail’ (loanword from English). See Chan, Lee & Yip (2022) and Lee & Yip (2024b) for a thorough discussion.

- (i) Zikzi lukdim zi **fan-faan-zoek**.
 until 6.o’clock then sleep-again-PC
 ‘(I) couldn’t sleep again until 6 am.’ (Forum, 2018-11-16)
- (ii) a. laai-coeng-**faan**/ laai-**faan**-coeng (Verb + RVC)
 pull-long-again pull-again-long
 ‘lengthen again’ (Chan, Lee & Yip 2022:26)
- b. feilou-**zo**/ fei-**zo**-lou (Monomorphemic verb)
 fail-PFV fail-PFV-fail
 ‘failed’ (Chan, Lee & Yip 2022:26)

(85) -Faan may follow PCs

Jausi taai daaiseng zigei wui seng-jat-seng jau **fan-zoek-faan**.
 sometimes too loud self will wake.up.a.bit again sleep-PC-again
 ‘Sometimes (his snoring) is too loud and wakes him up; then he would fall asleep again.’ (Forum, 2020-18-28)

(86) -Gwo cannot follow or precede PCs

- a. *Bunje zaatseng soeng zoi {**fan-zoek-gwo/ fan-gwo-zoek**}.
 midnight wake.up want again sleep-PC-again/ sleep-again-PC
- b. Bunje zaatseng soeng zoi **fan-gwo**.
 midnight wake.up want again sleep-again
 ‘(I) has accidentally woke up at midnight and wants to sleep again.’

The ordering of *-faan* and *-gwo* with respect to other postverbal elements is summarized in (87). *-Faan* competes with aspectual suffixes in the same position and follows PCs, whereas *-gwo* occupies the same slot as PCs and precedes aspectual suffixes.⁴¹

(87) The verbal template in Cantonese⁴²

VERBAL ROOT < {PC, *-gwo*} < {Asp-suffix, *-faan*}

6 Again lowering under the split aspect approach

In this section, we show how ‘again’ lowering in Cantonese can be derived using the same ingredients we have adopted to account for Mandarin ‘again’ skipping, with a crucial additional ingredient, namely, the split aspect approach (Section 6.1). We will see how splitting aspect into an outer aspectual projection (AspP_{outer}) and an

⁴¹ Certain varieties of Mandarin also have a postverbal ‘again’-type element, *-hui* ‘again, (lit.) return’ (Anonymized p.c.). Like Cantonese *-gwo*, it does not allow for wide-scope reading across nonfinite clauses, as in (i). Curiously, it also patterns with *-gwo* in preceding an outer aspectual suffix such as perfective *-le*, as in (ii).

(i) (Context: Zhangsan always wanted to listen to this song but he never got the time to do so.)

#Jintian ta xiang [ting-**hui** zhe shou ge].
 today 3SG want listen-again this CLF song

#Narrow scope: ‘Today, he wants to again listen to this song.’

(Anonymized p.c.)

*Wide scope: ‘Today, he again wants to listen to this song.’

(ii) a. Zuijin chongxin du-**hui-le** guanyu meixue taolun de shu.
 recently again read-again-PFV related aesthetics discussion DE book
 ‘Recently, [I] reread a book on aesthetics.’

(Anonymized p.c., Blog, 2018)

b. Zuijin ting-**hui-le** yi shou hen langman de Yingwen gequ.
 recently listen-again-PFV one CLF very romantic DE English song
 ‘Recently, [I] heard a very romantic English song again.’

(Anonymized p.c., Facebook, 2015-10-6)

⁴² This template is simplified and does not include RVCs, which precede PCs, and a universal quantificational suffix *-saai*, which may follow aspectual suffixes (S.-W. Tang 2003, Yip 2020, 2025). The complete template is given below. See the Appendix for more discussion.

(i) VERBAL ROOT < RVCs < {PC, *-gwo*} < {Asp-suffix, *-faan*} < *-saai*

inner aspect aspectual projection (AspP_{inner}) in the syntax captures the scopal differences between *-faan* and *-gwo* (Section 6.2).

6.1 Splitting aspect

It has been argued by a number of authors that the clausal spine has more than one aspectual projection in Chinese (Gu 1993, 1995, Tsai 2008, Huang, Li & Li 2009, Sybesma 2017, Lu, Lipták & Sybesma 2019, Yip 2020, 2025, S.-W. Tang 2022, Lee & Pan 2024) and in other languages (MacDonald 2008, Travis 2010, *i.a.*). As a general dichotomy, situation aspect is often placed within vP as *inner aspect* and viewpoint aspect above vP as *outer aspect*.⁴³ We dub it as *the split aspect approach*.

The outer aspect is associated with preverbal aspectual markers (e.g., progressive (*zheng*)*zai*) and some aspectual suffixes (e.g., experiential *-guo*, perfective *-le*, and imperfective *-zhe*) (Smith 1991, 1994; Huang, Li & Li 2009:101-105; Soh 2014), whereas the inner aspect is associated with some other aspectual suffixes (e.g., continuative *-zhe*, cf. Section 7 and Tsai 2008; see also Sun & Cheung 2022 for telic *-le*) and phase complements (Sybesma 2017, Tsai 2008).⁴⁴ The approach is summarized in (88).

(88) The split aspect approach to Chinese languages

- a. [TP ... [AspP_{outer} **ASP_{outer}** [vP ... [AspP_{inner} **ASP_{inner}** [VP ...
- b. Elements associated with Asp_{outer}:
Mandarin: (*zheng*)*zai*_{PROG}, *you* ‘have’; *-guo*_{EXP}, *-le*_{PFV}, *-zhe*_{IPFV} etc.
Cantonese: *jau* ‘have’; *-gwo*_{EXP}, *-zo*_{PFV}, *-gan*_{PROG}, etc.
- c. Elements associated with Asp_{inner}:
Mandarin: phase markers *-le*_{telic}, *-zhe*_{CONT}, PCs like *-diao*, *-zhao*, etc.
Cantonese: *-zyu*_{CONT}, PCs like *-jyun*, *-hou*, *-zoek*, etc.

We propose that Cantonese postverbal ‘again’ has a similar split: *-faan* is associated with Asp_{outer} and *-gwo* is associated with Asp_{inner}. The evidence comes from suffix ordering: *-faan* occupies an outer slot, competing with outer aspectual suffixes, whereas *-gwo* occupies an inner slot, competing with PCs. According to the Mirror Principle (Baker 1985), the suffix order mirrors the hierarchical structure that feeds into LF scope:

- (89) a. VERBAL ROOT < {PC, *-gwo*} < {Asp-suffix, *-faan*} (suffix order)
- b. [Asp-suffix/*-faan* ... [PC/*-gwo* ... [VERBAL ROOT ... (LF scope)

Following the lexicalist hypothesis (Gu 1995, Huang, Li & Li 2009, S.-W. Tang 2010, 2015), verbal suffixes are base-generated in V⁰ with the stem but associate with corresponding functional projections via agreement

⁴³ There have been attempts to further split both aspects into more fine-grained projections (see, for example, Sybesma 2017 for inner aspect and Gu 1995 for outer aspect). For the purposes of this paper, it suffices to only posit two aspectual projections, Asp_{outer} and Asp_{inner}.

⁴⁴ Perfective *-le*, in the work by Tsai (2008) and Sybesma (2017) (dubbed as “realization”), is analyzed as an inner aspect lower than *-guo*. However, Gu (1995) argues that the Asp head associated with *-le* should be higher than *-guo* and have a wider scope, since *-le* must follow *-guo*, as in (i) (See also Yip 2020 for additional evidence from Yangchun Yue). We thus associate *-le* with Asp_{outer}.

(i) Ta yijing {kan-**guo-le**/ *kan-**le-guo**} nei-ben shu le. Mandarin
3SG already read-EXP-PFV read-PFV-EXP that-CLF book PRF
‘He has already read that book.’ (Gu 1995:52)

for interpretation at LF. Accordingly, *-faan* and *-gwo* are base-generated in the vP layer as part of V^0 . *-Faan*, just like experiential suffixes, agrees with the outer aspect head outside vP. *-Gwo*, in contrast, agrees with the inner aspect head inside vP.

(90) Postverbal ‘again’ in Cantonese: agreement with Asp

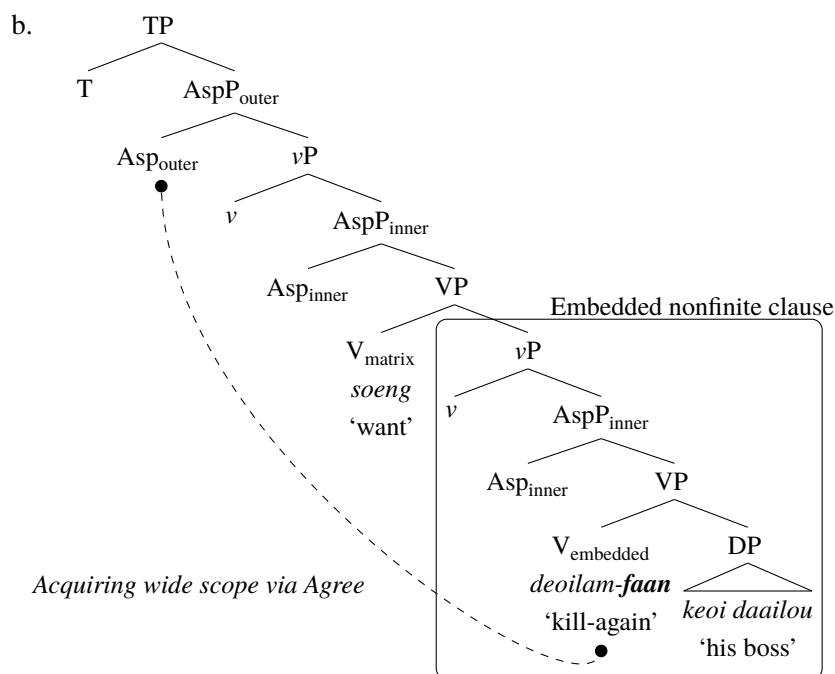
- a. ... [AspP_{outer} Asp_{outer} ... [vP ... [AspP_{inner} Asp_{inner} ... [VP V-*faan* ...
- b. ... [AspP_{outer} Asp_{outer} ... [vP ... [AspP_{inner} Asp_{inner} ... [VP V-*gwo* ...

6.2 ‘Again’ lowering as outer aspect agreement

With the ingredients in place, we can now derive *-faan* lowering cases such as the one in (91a) (reproduced from (72a)), when the embedded *-faan* takes wide scope. First, the matrix verb *soeng* ‘want’ takes a vP-sized nonfinite clause (i.e., restructuring clauses). Second, the verbal complex V-*faan* needs to agree with Asp_{outer}. Since the embedded clause is only vP and does not contain Asp_{outer}, V-*faan* agrees with the matrix Asp_{outer} and is interpreted there, which takes scope over the matrix VP headed by ‘want’. The structure is represented in (91b).

(91) Postverbal *-faan* ‘again’ taking wide scope through agreement

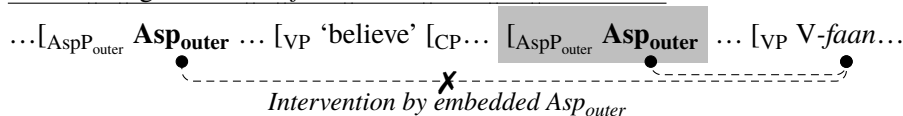
- a. Aaming *soeng* [deoilam-**faan** keoi daailou]. (again>want>kill)
 Ming want kill-again 3sg boss
 Wide scope: ‘Ming again wants to kill his boss.’



The explanation of why *-faan* lowering cannot occur across a finite clause boundary (e.g., over matrix verbs like *seon* ‘believe’) is very similar to the ones for ‘again’ skipping and aspect lowering. When the embedded clause is a CP, there is also an embedded Asp_{outer}. *-Faan* therefore agrees with the closest Asp_{outer} and is

interpreted in the embedded clause, giving rise to the narrow scope. That is, the embedded $\text{Asp}_{\text{outer}}$ intervenes between the matrix $\text{Asp}_{\text{outer}}$ and *-faan*.

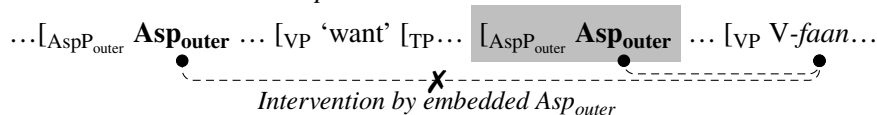
(92) Failure of agreement of *-faan* across finite CP boundaries



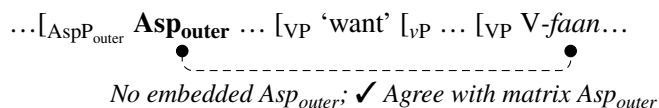
On the other hand, *-faan* generally allows for a narrow scope reading in nonfinite clauses, unlike aspectual suffixes when embedded under verbs like ‘want’/‘urge’ (but see Section 3). We suggest that in those cases, ‘want’ takes a TP which contains $\text{AspP}_{\text{outer}}$. Importantly, the aspectual requirement imposed on the embedded clause is compatible with *-faan*’s semantics (i.e., the ‘again’ presupposition). *-Faan* does not require [+D] at $\text{Asp}_{\text{outer}}$, as evidenced by its compatibility with stative verbs like *jau* ‘have’ (see (78) above).⁴⁵ Hence, *-faan* may agree with the embedded $\text{Asp}_{\text{outer}}$ to give a narrow scope reading. Indeed, *-faan* must agree with the embedded $\text{Asp}_{\text{outer}}$, since it is the closest Asp head. The wide scope reading can only be obtained when the embedded clause is small enough to *exclude* $\text{Asp}_{\text{outer}}$, i.e., a vP . In other words, the so-called scopal ambiguity of *-faan* is a result of different sizes of the nonfinite clauses, as schematized in (93).

(93) Scopal ambiguity of *-faan*

a. *Embedded TP: narrow scope*



b. *Embedded vP: wide scope*



We thus predict that when the nonfinite clauses contain materials indicating a TP size, *-faan* can only take narrow scope. This prediction is borne out in the examples below where the contexts favor a wide scope reading. With the semicomplementizer *waa* (see Yeung 2006, J. Huang 2021) in (94), adverb *dou* ‘also’ in (95), and internal *lin* ‘even’ focus fronting in (96), *-faan*’s scope is restricted to the embedded verb and results in infelicity. No wide scope readings over the matrix verb ‘urge’ are available. The patterns parallel what we have seen for aspect lowering in Section 3.1 and preverbal ‘again’ skipping in Section 4.2.

(94) *-Faan* cannot take wide scope with semicomplementizer *waa*

[Context: I used to urge Ming to apply for PhD. Yet, he kept ignoring me and even became mad at me, so I just gave up. Today, our teacher and I see that the government has additional funding for PhD studies, ...]

#Ngo hyun Aaming [*waa* bou-**faan** PhD].

1sg urge Ming COMP apply-again PhD

#Narrow scope: ‘#I urge Ming to reapply for PhD.’

*Wide scope: ‘I again urge Ming to reapply for PhD.’

⁴⁵ Other stative verbs *-faan* may attach to include *hai* ‘be at (somewhere)’, *dak* ‘only have’, *dangjyu* ‘equal’, *ci* ‘resemble’, etc., which are all incompatible with perfective *-zo*. That said, *-faan* cannot attach to modals and some other stative verbs like ‘want’ (see (74) above) for reasons unknown to us.

(95) *-Faan* cannot take wide scope with adverb *dou* ‘also’

[Context: Same as (94), except that Fan already applied for PhD today.]

#Ngo hyun Aaming [*dou* bou-**faan** PhD].

1sg urge Ming also apply-again PhD

#Narrow scope: ‘#I urge Ming to also reapply for PhD.’

*Wide scope: ‘I again urge Ming to also apply for PhD.’

(96) *-Faan* cannot take wide scope with internal *lin* ‘even’ focus fronting

[Context: Same as (94), except that what I urged Ming to apply included master studies, and that there is also funding for master studies.]

#Ngo hyun Aaming [*lin* *master* *dou* bou-**faan**].

1sg urge Ming even master DOU apply-again

#Narrow scope: ‘#I urge Ming to reapply even for master studies.’

*Wide scope: ‘I again urge Ming to apply even for master studies.’

We have seen that *-faan*’s scope is regulated by whether $\text{Asp}_{\text{outer}}$ is embedded. Another postverbal ‘again’-type element, *-gwo*, lacks a wide scope reading altogether when embedded (i.e., *-gwo* lowering is impossible). Unlike *-faan*, *-gwo* agrees with $\text{Asp}_{\text{inner}}$. The absence of wide scope is a natural consequence of the current proposal. To elaborate, the nonfinite clause, even as a vP , always contains $\text{Asp}_{\text{inner}}$. Thus, *-gwo* always agrees with the embedded $\text{Asp}_{\text{inner}}$ and cannot agree with the matrix one to get wide scope. An illustration is given below for a sample example (97a) (reproduced from (72b)).

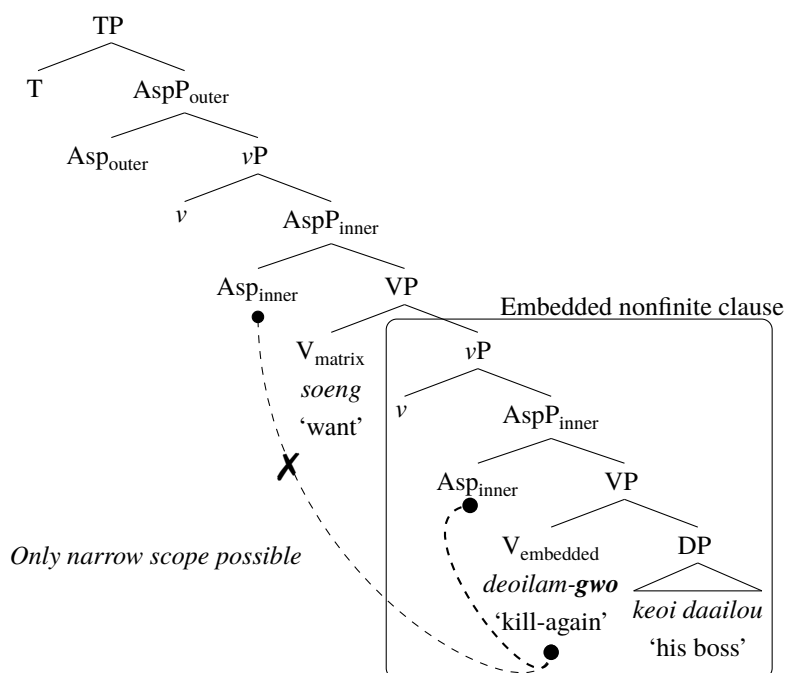
(97) Postverbal *-gwo* ‘again’ lacks wide scopea. #Aaming soeng [*deoilam-gwo* keoi daailou].

(want>again>kill)

Ming want kill-again 3sg boss

Narrow scope only: ‘#Ming wants to again kill his boss.’

b.



Extending this split aspect approach back to Mandarin preverbal ‘again’ skipping, we can now identify the AspP that *you* moves to as AspP_{outer} above vP. A question arises as to whether *zai* also associates with Asp_{inner} within vP, or stays at the lowest VP. We believe that the former is the case. Witness (98), where *zai* takes wide scope over the PC, achievement marker *diao*. The reading is that “I” would again achieve the selling of Minecraft, but not that “I” would achieve the reselling of Minecraft.

- (98) (Context: Markus Persson, the founder of Minecraft, is sharing his voidness after selling Minecraft away to Microsoft.)

... Wo hai shi na-ge reng hui **zai** mai-diao Minecraft de hundan eryi.
 1SG still COP that-CL still will again sell-PC Minecraft DE jerk only
 ‘...I’m still that jerk who would again sell Minecraft away.’ (again>achievement)
 (News, 2015-8-31)

Therefore, we obtain the following associated (moved) positions of *you* (AspP_{outer}) and *zai* (AspP_{inner}) in (99). This parallels Cantonese where *-faan* agrees with Asp_{outer} and *-gwo* agrees with Asp_{inner}.

- (99) ... [AspP_{outer} (*you*) [Asp_{outer}... [vP... [AspP_{inner} (*zai*) [Asp_{inner}... [VP V...

The reason why *zai* does not allow for ‘again’ skipping now follows directly from intervention by AspP_{inner} in vP which we have proposed for *-gwo*: when *zai* is embedded in a vP/restructuring clause, the embedded AspP_{inner} is always the closest landing site as compared to the matrix one. ⁴⁶

We close this section by pointing out that the account of the asymmetry between *you/-faan* and *zai/-gwo* relies on the fact that nonfinite clauses contain inner aspect, such that there is always intervention to *zai*’s movement to matrix AspP_{inner} and *-gwo*’s agreement with matrix Asp_{inner}. We argue that this case indeed provides important evidence that there is a *minimal size* of nonfinite clauses: they must contain the inner aspect and at least be vP. As will be discussed in the next section, this offers an explanation of why aspect lowering does not apply to all aspectual suffixes as well as phase complements, an observation that has largely gone unnoticed.

7 The minimal size of nonfinite clauses

We have seen how a smaller size of nonfinite clauses, namely vP, is supported by the presence of *-faan* lowering. There is however little discussion on the *minimal* size of nonfinite complement clauses. While it is usually

⁴⁶ To further clarify our position, we hold that the interpretation site for ‘again’-type elements is always their merge position. Both *you* and *zai* can merge below inner aspect, but while *you* may move to the outer aspect projection, *zai* cannot and is restricted to movement to the inner aspect projection, a difference arising from their different relationships with different aspect heads. An example where *zai* scopes below inner aspect is given below:

- (i) (Zhangsan and Lisi were good friends who shared snacks all the time, but they did not like sharing them with anyone else. One day, they bought some snacks, and Zhangsan tried to hide them in a closet, but he did not hide them very well; half of the snacks were still visible from outside. When Lisi saw this, he wanted to hide them again so that they were not visible from outside.)

Lisi xiang zai ba lingshi cang-hao.
 Lisi want again DISP snack hide-COMPL
 ‘Lisi wants to hide the snacks again so they are well hidden.’

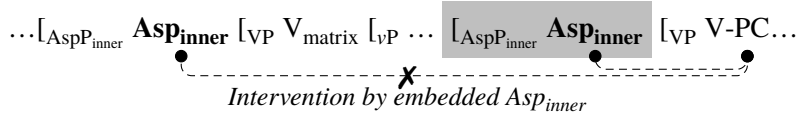
In this case, *zai* is interpreted as modifying *cang* ‘hide’ rather than *cang-hao* ‘hide-COMPL’, as the completive aspect does not hold in the antecedent event.

assumed to be *vP* (as opposed to *VP*) (Grano 2012:§5, Grano 2014, Grano 2015:§6; N. Huang 2018, C.-T. J. Huang 2022), there has been, as far as we know, no evidence offered for this claim. We argue that the *absence* of *-gwo* lowering supports that *vP* is the *minimal* size of nonfinite complement clauses. The clauses must at least contain inner aspect; otherwise, one would expect *-gwo* to agree with matrix *Asp_{inner}* for wide scope. Thus, we obtain the fined-grained typology of complement clauses in (100).

- (100) Complement clause sizes, final version (updated from (44))
- a. $V [CP \dots [TP \dots [AspP_{outer} \dots [vP \dots [AspP_{inner} \dots [VP \dots]]]]]]$ (Type I, finite)
 - b. $V [TP \dots [AspP_{outer} \dots [vP \dots [AspP_{inner} \dots [VP \dots]]]]]]$ (Type II, nonfinite)
 - c. $V [vP \dots [AspP_{inner} \dots [VP \dots]]]]$ (Type III, nonfinite)
 - d. $*V [VP \dots]$ (Unattested)

Further evidence comes from other inner aspectual elements like phase complements. Since PCs are also associated with *Asp_{inner}*, we predict that they cannot take wide scope when embedded (i.e., PC lowering should be impossible), if nonfinite clauses must at least contain *AspP_{inner}*, as shown in (101).

- (101) Prediction: No PC lowering



This prediction is borne out. As shown in (102), the achievement markers *-diao* and *-hou* attach to the embedded verb and must be interpreted there. The matrix verbs do *not* obtain the same achievement reading — there lacks a reading where the urging/forcing events have been achieved. The matrix verbs may even take an outer aspect incompatible with achievement, such as progressive and/or imperfective aspects (indicated by *zai...-zhe* and *haidou...-gan*).

- (102) Phase complements cannot take wide scope when embedded

- a. Wo (zai) quan(-zhe) ta chi-**diao** zhe-wan fan. [Mandarin]
 1SG PROG urge-IPFV 3SG eat-PC this-CLF rice
 ‘I urge/am urging him to eat up this bowl of rice.’
 NOT: ‘I finished urging him to eat this bowl of rice.’
- b. Ngo (haidou) bik(-gan) keoi so-**hou** ni dou mun [Cantonese]
 1SG PROG force-PROG 3SG lock-PC this CLF door
 ‘I force/am forcing him to lock the door properly.’
 NOT: ‘I finished forcing him to lock the door.’

The incompatibility of PCs *-diao* and *-hou* with progressive and/or imperfective aspects is supported by (103) without embedding.

- (103) a. *Ta zai **chi-diao-zhe** zhe-wan fan. [Mandarin]
 3SG PROG eat-PC-IPFV this-CLF rice
 Int.: ‘S/he is eating up this bowl of rice.’
- b. *Ngo haidou **so-hou-gan** ni dou mun [Cantonese]
 1SG PROG lock-PC-PROG this CLF door
 Int.: ‘I’m locking the door properly.’

This explanation extends to the verbal suffix *-zhe* in Mandarin. To begin with, *-zhe* has two uses. The first one serves as an imperfective aspect, which may be durative (stative) or progressive (dynamic) (Smith 1991, 1994, Soh 2014). It is compatible with the preverbal aspectual marker *zai* and sentence-final particle *ne*, both expressing progressive aspect, as in (104a). It is thus reasonable to treat imperfective *-zhe* as associating with outer aspect. The second use is dubbed as a phase marker by Tsai (2008), which only attaches to verbs with a sense of “attachment”, such as verbs of posture like *zuo* ‘sit’ and verbs of carrying like *bao* ‘hold’, among others (see also Li & Thompson 1981, Yuan 1993), as in (104b). This use has been analyzed as a type of inner aspect (Tsai 2008:683; Grano 2014:23n.8). For expository purposes, we label it as a continuative aspect (i.e., indicating that a state is continuing).⁴⁷

(104) The outer-inner split of Mandarin aspectual suffix *-zhe*

- a. [Zhangsan *zai* **da-zhe** Lisi] *ne*. (imperfective, Asp_{outer})
 Zhangsan PROG hit-IPFV Lisi SFP
 ‘Zhangsan is hitting Lisi.’ (Li & Thompson 1981:219)
- b. *Ta* *zai* *fangzi-li* **zuo-zhe**. (continuative, Asp_{inner})
 3SG at house-inside sit-CONT
 ‘S/he is sitting in the house.’ (Li & Thompson 1981:219)

Tsai (2008), citing Yuan (1993), points out that only inner aspectual *-zhe*_{CONT} can be used in imperatives, whereas outer aspectual *-zhe*_{IPFV} cannot, as shown in (105).

- (105) a. *[(*Zai*) **da-zhe** Lisi] (*ne*)! (imperfective, Asp_{outer})
 PROG hit-IPFV Lisi SFP
 Int.: ‘Hit Lisi!’/‘To be hitting Lisi!’
- b. *Zai* *fangzi-li* **zuo-zhe**! (continuative, Asp_{inner})
 at house-inside sit-CONT
 ‘Sit in the house!’

Other outer and inner aspectual elements show similar contrasts. Experiential aspect *-guo* is banned in imperatives, whereas achievement marker PC *-diao* is not, as in (106).⁴⁸ The pattern in (105) thus supports that imperfective *-zhe* is an outer aspect and continuative *-zhe* is an inner aspect.

⁴⁷ The aspectual contribution by continuative *-zhe* is quite similar to PC *-zhu*, as in *zhan-zhe* ‘standing’ and *zhan-zhu* ‘standing (still)’, both indicating an ongoing state. However, *-zhe* is a verbal suffix whereas *-zhu* is a verbal complement, as evidenced by the contrast in *-de/bu-* ‘able/not’ infixation:

(i) **zhan-de/bu-zhe* vs. ^{OK}*zhan-de/bu-zhu*
 stand-able/not-CONT stand-able/not-PC
 Int.: ‘to be (in)able of standing still’

⁴⁸ Note that the verbal suffix *-le* can be used in imperatives, but it is argued to be a telic use associated with Asp_{inner}, rather than a perfective use (Sun & Cheung 2022).

- (106) a. *Chi-**guo** zhe-wan fan! (experiential, Asp_{outer})
 eat-EXP this-CLF rice
 Int.: ‘To have an experience of eating this bowl of rice!’
- b. Chi-**diao** zhe-wan fan! (achievement PC, Asp_{inner})
 eat-PC this-CLF rice
 ‘Eat up this bowl of rice!’

With the outer-inner split of *-zhe* in place, we can now test aspect lowering. The prediction is that only the outer *-zhe*_{IPFV} can take wide scope when embedded under nonfinite clauses, but not the inner *-zhe*_{CONT}. This prediction is borne out. Grano (2014) observes a case of *-zhe*_{IPFV} lowering in (107a), where *-zhe*_{IPFV} takes scope over the matrix verb *wei* ‘feed’. We further observe that *-zhe*_{IPFV} does not need to be interpreted locally, as evidenced by the felicitous continuation in (107b) that the eating event does not occur. This is thus a genuine case of aspect lowering.

- (107) Outer aspectual *-zhe*_{IPFV} may take wide scope when embedded
- a. Wo (zai) wei ta [chi-**zhe** fan] ne.
 1SG PROG feed 3SG eat-IPFV fan SFP
 ‘I am feeding him/her a meal.’ (Grano 2014:23)
- b. ... Ke ta nao piqi, yi-kou dou mei chi.
 but 3SG have temper one-bite also NEG.PFV eat
 ‘... but s/he is having a temper and doesn’t even eat one bite of it.’

We provide one more example of *-zhe*_{IPFV} lowering in (108). Likewise, the embedded *-zhe*_{IPFV} takes matrix scope over ‘force’ rather than embedded scope over ‘sing’.⁴⁹

- (108) [Context: Ming never sings in Karaoke. You really want to hear his voice and are forcing him to sing. However, Lisi keeps distracting you and you still haven’t got Ming singing. You say to Lisi:]
 Wo (zheng) bi ta [chang-**zhe** ge] ne! Ni bie daoluan!
 1SG PROG force 3SG sing-IPFV song SFP 2SG NEG casue.trouble
 ‘I’m now forcing him to sing! Don’t cause trouble!’

On the other hand, the inner aspectual *-zhe*_{CONT} cannot acquire wide scope across nonfinite clauses. Example (109a) lacks a reading where the forcing event/state is continuative. Furthermore, the matrix verb may take

⁴⁹ Like Mandarin *-zhe*_{IPFV}, Cantonese progressive *-gan* exhibits aspect lowering, as in (1). Aspect lowering is general to all the outer aspectual suffixes. See, for example, C.-T. J. Huang (2022: .19n2) for perfective *-le* in Mandarin, Yip (2024) for exclusive suffix *-dak* ‘only’ in Cantonese, and Francis & Matthews (2006) and T. T.-M. Lee (2024a) for similar lowering phenomena in coverb and serial verb constructions. We thank an anonymous reviewer for raising this point.

- (i) [Context: I’m trying to feed a baby milk, but the baby keeps crying and doesn’t drink the milk because you are making noise next to him. I say to you:]
 Zing-di! Ngo (jigaa) wai keoi [jam-**gan** naai] wo3!
 quiet-bit 1SG now feed 3SG drink-PROG milk SFP
 ‘I’m feeding him (to drink) milk!’

the outer aspect experiential *-guo*, which is incompatible with *-zhe_{CONT}* as in (109b), showing that *-zhe_{CONT}* in (109a) can only be interpreted in the embedded clause.⁵⁰

(109) Inner aspectual *-zhe_{CONT}* cannot take wide scope when embedded

- a. Wo bi(-guo) ta [zai fangzi-li zuo-**zhe**].
 1SG force-EXP 3SG at house-inside sit-CONT
 ‘I force/have forced him/her to sit still in the house.’
 NOT: ‘I continue forcing him/her to sit in the house.’
- b. *Ta zai fangzi-li zuo-**zhe**-guo.
 3SG at house-inside sit-CONT-EXP
 Int.: ‘S/he have been sitting still in the house.’

The lack of *zhe_{CONT}*-lowering is further supported by (110a). The matrix verb takes the inner aspect com-
 pletive *-wan* that is incompatible with *-zhe_{CONT}* independently as in (110b). Thus, *-zhe_{CONT}* in (110a) lacks
 matrix scope.

- (110) a. Wo quan-wan ZS [zai fangzi-li zuo-**zhe**], jiu quan LS bie zhan-zhe.
 1SG urge-PC ZS at house-inside sit-CONT, then urge LS NEG.PROH stand-CONT
 ‘After I finished urging Zhangsan to sit still in the house, I urge Lisi not to stand still.’
- b. *Zhangsan zai fangzi-li zuo-**zhe**-wan.
 Zhangsan at house-inside sit-CONT-PC
 Int.: ‘Zhangsan finished sitting still in the house.’

The asymmetries in lowering between outer aspectual *-zhe_{IPFV}* and inner aspectual *zhe_{CONT}* further support
 that nonfinite clauses are as least as large as *vP*. A restructuring clause does not contain *Asp_{outer}* and thus allows
-zhe_{IPFV} to agree with matrix *Asp_{outer}*, but it contains *Asp_{inner}* that triggers intervention to block *-zhe_{CONT}* from
 matrix agreement, as in (111).

(111) Asymmetries in aspect lowering between *-zhe_{IPFV}* and *-zhe_{CONT}*

- a. [Asp_{outer} Asp_{outer} [vP ... [Asp_{inner} ... [vP V [vP ... [Asp_{inner} ... [vP V -zhe_{IPFV} ...
 ✓ Agreement, no intervention
- b. [Asp_{inner} Asp_{inner} [vP V [vP ... [Asp_{inner} Asp_{inner} [vP V -zhe_{CONT} ...
 ✗ Intervention by embedded Asp_{inner}

⁵⁰ Grano (2014) cites (i) as a case of *-zhe_{CONT}* failing to take wide scope, since the matrix clause has perfective negation *meiyou* ‘haven’t’. However, it is far from clear that this case shows the failure of lowering, since *-zhe_{CONT}* is compatible with *meiyou*, as in (ii).

- (i) Wo meiyou jiao ta [kao-**zhe** qiang].
 1SG NEG.PFV ask 3SG lean-CONT wall
 ‘I did not ask him to lean against the wall.’

(Y. Huang 1994:29, cited by Grano 2014:23n.8)

- (ii) Wo meiyou kao-**zhe** qiang.
 1SG NEG.PFV lean-CONT wall
 ‘I did not lean against the wall.’

8 Conclusion

To conclude, we have argued for a size-based view on finiteness in Chinese languages under the split aspect approach. Evidence is drawn from the scopal behavior of various ‘again’ and aspectual elements in Mandarin and Cantonese, most of which is first observed in this paper.

There are two key findings. First, preverbal *you* ‘again’ in Mandarin, postverbal *-faan* ‘again’ in Cantonese, and aspectual suffixes in both languages, which are all associated with outer aspect, may take exceptional scope that mismatches with the pronounced position: embedded scope for matrix *you* (“skipping”), matrix scope for embedded *-faan* and suffixes (“lowering”). Moreover, elements associated with inner aspect do not allow skipping or lowering across nonfinite clause boundaries, such as Cantonese *-gwo* ‘again’ and phase complements, as well as Mandarin *zai* ‘again’. Second, *you* skipping and *-faan*/aspect lowering are sensitive to a certain size of nonfinite clause, namely, vP (restructuring clauses). A TP-sized nonfinite clause and a CP-sized finite clause render skipping and lowering impossible. The empirical patterns are summarized in Table 5.

Table 5 Different nonfinite clause sizes and the availability of aspectual/‘again’ exceptional scope phenomena, final

	Size	Asp lowering <i>-faan</i> lowering	PC lowering <i>-gwo</i> lowering	<i>you</i> skipping	<i>zai</i> skipping
Associated with		Asp _{outer}	Asp _{inner}	Spec, AspP _{outer}	Spec, AspP _{inner}
Under ‘believe/say’	CP	✗	✗	✗	✗
Semi-COMP <i>shuo/waa</i>	TP	✗	✗	✗	✗
Adv <i>ye/dou</i> ‘also’	TP	✗	✗	✗	✗
Internal focus	TP	✗	✗	✗	✗
None of the above	vP	✓	✗	✓	✗

We have proposed an account that draws on (i) variable sizes of nonfinite clauses with a lower bound at vP (N. Huang 2018, C.-T. J. Huang 2022; cf. Wurmbrand & Lohninger 2023); (ii) split aspect projections with AspP_{outer} (above vP) and AspP_{inner} (below vP) (e.g., Tsai 2008, Sybesma 2017); and (iii) association of ‘again’ and aspectual elements with relevant Asp projections, analyzed as movement for preverbal *you* and agreement for other postverbal elements (building on Liu 2021, 2024 and Grano 2014, N. Huang 2018, respectively). The exceptional scope is a by-product of this association: reconstruction of *you* generates embedded scope, and agreement of *-faan*/aspectual suffixes yields matrix scope. Importantly, such association is not unrestricted: it is disrupted when the embedded clause is larger than vP (e.g., TP), which we modeled as a result of (defective) intervention by embedded Asp_{outer}.

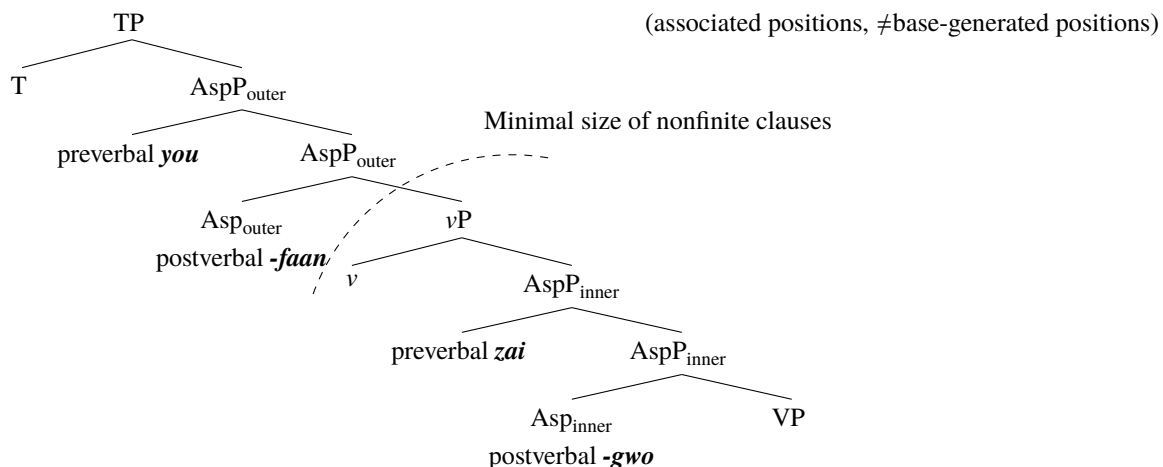
The empirical findings and proposed account have several important consequences on (i) finiteness and complementation and (ii) the cartography of ‘again’ and aspectual elements. First, finiteness distinction is structurally encoded in Chinese languages, but it is not a simple dichotomy with [$\pm$ finite] or [$\pm$ tense]. Rather, it is a *gradient* notion manifested structurally as a hierarchy of complement clause sizes, as evidenced by the blocking effects of nonfinite TP to ‘again’ skipping and ‘again’/aspect lowering. This perspective not only aligns perfectly with the reduced clause size analysis of restructuring and its related phenomena but also

substantially supports Wurmbrand & Lohninger (2023)’s view on complementation (originally circulated in 2020) that is recently adopted by He (2024) and C.-T. J. Huang (2022) to Chinese.

Moreover, there is a *minimal* size of nonfinite clauses, namely, vP. While this assumption has been adopted widely (Grano 2014, N. Huang 2018, C.-T. J. Huang 2022), this study is the first to provide substantial evidence from the systematic absence of skipping/lowering phenomena of elements associated with Asp_{inner}.

Second, there is a fined-grained structural distinction of ‘again’ and aspectual elements on the clausal spine, the most crucial one being the relative position with respect to vP (e.g., Tsai 2008, Sybesma 2017, Lu, Lipták & Sybesma 2019; see Lin & Liu (2009) for ‘again’), as represented in (112).

(112) The cartography of ‘again’ elements in Chinese languages



The distinction plays a significant role in determining the availability of skipping/lowering: only elements associating above vP allows exceptional scope across nonfinite clauses. On a deeper level, the dichotomy reflects two of three domains in cartographic syntax (Cinque 1999, Rizzi 1997) — the inflectional domain and lexical domain (besides the complementizer domain), which has gained strong support from various empirical phenomena in Chinese (e.g., modals, light verbs, applicatives; see Tsai 2015). It also offers a potential explanation for why nonfinite clauses cannot be smaller than vP: vP is the upper bound of the lexical domain and the lower bound of the inflectional domain. It is hoped that this perspective may further our understanding of both clausal architecture and finiteness in Chinese languages and beyond.

A Appendix: Cross-type ambiguity in clause-taking predicates

A reviewer pointed out an apparent inconsistency of the current analysis with the syntax-semantics mapping in the ICH when we look at predicates like *dasuan* ‘plan’, which has been argued to be obligatorily “situation”-taking but allows vP complements in our proposal. In response to this apparent challenge, we pursue a line that questions the implicit assumption that “predicates that are glossed as ‘plan’ would always be situation/TP-taking across languages.” In essence, we claim that predicates like *dasuan* ‘plan’ exhibit ambiguity in Chinese, which may take either a type II (TP) or a type III (vP) complement, following the suggestion of C.-T. J. Huang (2022: 24, fn. 8) that “many predicates that may be listed under one type may belong to another type.” To first provide a comprehensive picture, we present our full suite of diagnostics for “situation” TPs and “event” vPs.

- (113) a. “Situation” complements are necessarily TPs and not vPs. They uniquely allow certain modals or aspects, such as *yao*.

- b. Aspectless “event” complements are necessarily vPs and not TPs. They uniquely allow aspect lowering, ‘again’ skipping and ‘again’ lowering.
- c. (Other TP-level materials, such as the semicomplementizer *shuo*, the adverb *ye* ‘also’, and internal foci, are irrelevant here, since TPs can be mapped either from a situation or from an event.)

Some predicates satisfy both sets of diagnostics, while others satisfy only one. We show that *dasuan* belongs to the group that supports both, reflecting an ambiguity. Our argument unfolds in several steps. We begin by illustrating predicates that are unambiguously either “situation”-taking (TP) or “event”-taking (vP). We then turn to those we consider ambiguous (like *dasuan*) and attempt to tease apart their semantic interpretations. Finally, we offer some remarks on the distinction between the two types in partial and exhaustive control.

An example of predicates that are unambiguously “event”-taking is *changshi* ‘try’ (114). An example of predicates that are unambiguously “situation”-taking is *zhengqu* ‘strive’ (115).

(114) ‘Try’

- a. *TP diagnostic*: ✗
 Zhangsan changshi [(*)yao (*)mingtian) xi dama].
 Zhangsan try want tomorrow smoke marijuana
 ‘Zhangsan tried smoking marijuana.’ (C.-T. J. Huang 2022:25)
- b. *vP diagnostic*: ✓
 Zhangsan changshi [jie-guo yan].
 Zhangsan try quit-EXP smoking
 ‘Zhangsan has tried quitting smoking before.’ (C.-T. J. Huang 2022:25)

(115) ‘Strive’

- a. *TP diagnostic*: ✓
 Zhangsan zhengqu [yao hui Xianggang].
 Zhangsan strive want return Hong.Kong
 ‘Zhangsan strives to return to Hong Kong.’
- b. *vP diagnostic*: ✗
 Zhangsan zhengqu(-guo) [chuban(*)-guo) shi ben shu].
 Zhangsan strive-EXP publish-EXP ten CLF book
 ‘Zhangsan has strived to publish ten books.’

In contrast, predicates like *quan* ‘urge’ pass both sets of diagnostics and is thus ambiguous between TP-selecting and vP-selecting usages:

(116) ‘Persuade’

- a. *TP diagnostic*: ✓
 Zhangsan quan Lisi [yao xiaoxin].
 Zhangsan persuade Lisi want careful
 ‘Zhangsan persuaded Lisi to be careful.’

b. *vP diagnostic*: ✓

Zhangsan quan Lisi [jie-guo yan].
 Zhangsan persuade Lisi quit-EXP smoking
 ‘Zhangsan has persuaded Lisi to quit smoking before.’

Before the phenomena reported in this paper come to light, these patterns have two plausible accounts. One is that aspect lowering is not a reliable vP diagnostic and can extend to TP complements under restricted conditions, perhaps with certain semantic properties. The other is that the applicability of TP and vP diagnostics is fully regular, and only some predicates are ambiguous in the clauses that they may select. The discovery of ‘again’ skipping strongly supports the second view, as it aligns closely with aspect lowering and serves as an additional vP diagnostic. The *corelation* between different dependencies calls for a common analytic component: the syntactic relation with AspP. If we abandon this analysis and pursue the first option, it would be a complete accident that ‘again’ skipping and aspect lowering somehow can apply across some TPs but not in others. Examples of unambiguously type III predicates that allow ‘again’ skipping are reported in the manuscript, and below is an example of an unambiguously type II predicate that does not:

- (117) (Aaming, an international student, is from Hong Kong. He went back to Hong Kong once last year. He didn’t need to fight for the opportunity; after all, he is a Hongkonger, so if he wanted to go back, he simply went. However, this year, he ended up being wanted by the government, which made it impossible for him to return. Still, he’s homesick and desperately wants to go back. So ...)

#Aaming you zhengqu hui Xianggang.
 Ming again strive return Hong.Kong
 Intended: ‘Ming strives to again to return to Hong Kong.’

It is thus not unnatural to consider *dasuan* a predicate that is ambiguous between TP-selecting and vP-selecting, just like *quan*.

An immediate question is whether the proposed ambiguity between TP-selecting vs. vP-selecting can be semantically distinguished. Our answer is that it apparently can. This distinction is difficult to isolate with future-oriented predicates like *dasuan* or *quan*, but with *wangji* ‘forget’, we are able to identify a context in which a minimal pair shows a clear contrast in felicity:

(118) a. *TP with TP diagnostic*: OK

Wo wangji(-le) [yao guan zhe shan men], danshi suiran wo wangji-le, wo haishi
 1SG forget-PFV want close this CLF door but although 1SG forget-PFV 1SG still
 guan-le.
 close-PFV
 ‘I forgot that I need to close this door, but although I forgot, I still closed it.’

b. *vP with vP diagnostic*: #

#Wo wangji [guan-guo zhe shan men], danshi suiran wo wangji-le, wo haishi guan-le.
 1SG forget close-EXP this CLF door but although 1SG forget-PFV 1SG still close-PFV
 ‘I’ve forgotten to close this door, but although I forgot, I still closed it.’

In fact, the sequence in (118b) itself has two possible readings, one aligning with vP and the other aligning with TP. The reading in (118b) represents the one aligning with vP. The other is illustrated below:

(119) *TP with embedded aspect: Also OK*

Wo wangji(-le) [(ziji) guan-guo zhe shan men], danshi suiran wo wangji-le, wo haishi
 1SG forget-PFV self close-EXP this CLF door but although 1SG forget-PFV 1SG still
 guan-le.
 close-PFV

‘I forgot that I’ve closed this door, but although I forgot, I still closed it.’

Examples (118a) and (119) show that having a TP complement unlocks a layer of interpretation that is impossible with a vP complement by introducing a time point outside the event under discussion. On par with the ICH, we consider this the root of the ambiguity between types II and III, and we consider *wangji* an example that carries this type of ambiguity. Likewise, we argue that *dasuan* carries a similar ambiguity, though we acknowledge that it is difficult to tease apart the exact interpretation corresponding to each usage, due to the inherent future orientation in its semantics.

Further support comes from exhaustive control (EC) vs. partial control (PC) (D. Li 2024). Among type II and type III predicates, some exhibit EC, while others PC:

(120) a. *EC verb*

Lisi shefa liu dian chufa /*jihe.
 Lisi try six o’clock set.off gather
 ‘Lisi tries to set off/*gather at six o’clock.’

b. *PC verb*

Lisi dasuan liu dian chufa /jihe.
 Lisi plan six o’clock set.off gather
 ‘Lisi plans to set off/gather at six o’clock.’

(D. Li 2024: 335)

If the possibility of PC is associated with TP complements, a clear contrast can again be made between type II and (unambiguously) type III complements. In this sense, (120a) represents a type III predicate, and (120b) represents a type II predicate. This argument is further strengthened by the observation that PC is blocked when we apply vP diagnostics on a PC verb (i.e., type II predicate):

(121) a. Lisi dasuan-guo liu dian jihe.

Lisi plan-EXP six o’clock gather
 ‘Lisi has planned to gather at six o’clock.’

b. *Lisi dasuan liu dian jihe-guo.

Lisi plan six o’clock gather-EXP
 Intended: ‘Lisi has planned to gather at six o’clock.’

The aforementioned points motivate us to stick to an account where predicates like *dasuan* can be semantically ambiguous between type II and type III usages.

B Appendix: Inner aspect and phase complements

Most attested postverbal inner aspectual elements are “phase complements” (PCs). This term is first introduced by Chao (1965). PCs refer to postverbal elements expressing “the phase of an action in the first verbs rather than

some result in the action or goal ” (p.446, Sect. 6.6.3). While they often share similar grammatical properties with resultative verbal complements (RVCs), their semantic contribution is aspectual.

PCs may follow resultative verbal complement (RVCs) and should be distinguished from them. The ordering suggests that PCs are structurally higher than RVCs.

(122) Phase complements following RVCs

- a. haumei ngodei tung Hong Waa zungjyu **daa-hoi-dou** waatai, faatjin jyunloi
 afterwards 1PL with Akina Hong eventually hit-open-ACHV topic discover actually
 keoi hou funny tung nice.
 3SG very funny and nice
 ‘We eventually opened up the topic with Akina Hong, and found that she is really funny and nice.’
 (News, 2022-1-22) [C]
- b. Zenme ba shouji nong-huai-diao erqie kan-bu-chulai shi renweide?
 how BA cell.phone make-broken-ACHV and see-not-out COP man.made
 ‘How do you break a cell phone without letting people know that it was you?’ (Forum,
 2021-12-26) [M]

B.1 Inner aspect in Mandarin

Examples of inner aspectual elements in Mandarin are given below. (a-g) refer to the endpoint of events, expressing meanings of event achievement, completion, successful perception, etc. (h-i) make reference to the internal intervals of events and express continuation.

(123) Postverbal inner aspectual elements in Mandarin

(Chao 1965:446-450, Li & Thompson 1981:65-66, Sybesma 2017, Lu, Lipták & Sybesma 2019)

- a. *zhao* ‘be on target’, e.g. *zhao-zhao* ‘found’
- b. *dao* ‘arrive’, e.g. *mai-dào* ‘bought’
- c. *diao* ‘’, e.g. *mai-diao* ‘sold’
- d. *cheng* ‘succeed’, e.g. *kan-cheng* (*dian ying*) ‘succeeded in seeing movies’
- e. *hao* ‘good’, e.g. *suo-hao* ‘locked properly’
- f. *wan* ‘finish’, e.g. *chi-wan* ‘finished eating’
- g. *jian* ‘see (vs. look)’, e.g. *tīng-jian* ‘hear (vs. listen)’
- h. *zhu* ‘hold’, e.g. *zhan-zhu* ‘stand still’
- i. *zhe* ‘IPFV’, e.g. *zuo-zhe* ‘sit still’ (verbs with degree of attachment, Tsai 2008)

These elements are regarded as a middle aspect layer in Sybesma (2017) and Lu, Lipták & Sybesma (2019). The top layer is occupied by the aspectual suffix *-le*, and the lowest layer is occupied by resultative verbal complements (RVCs). The three aspect layers are all in between vP and VP.

(124) [_{vP}...[_{AspP3-Realization} *-le* [_{AspP2} PCs [_{AspP1-Telicity} RVCs [_{VP}...

(125) Relative order of phase complements with other postverbal elements

- a. $PCs < le$ (Sybesma 2017:303)
 Women mai-**diao-le** fangzi.
 1PL sell-PC-PFV house
 ‘We finally sold the house.’
- b. $RVCs < PCs < le$ (Lu, Lipták & Sybesma 2019:303)
 ta bb wo-de yundongxie pao-**huai-diao-le**
 3SG BA 1SG-SUB sport.shoe run-broken-PC-PFV
 ‘He ran my running shoes completely to pieces’

In Tsai (2008), PCs are regarded as the lowest layer of aspectual projections (Inner Aspect), including *zhe*² ‘keep/still’ (attach to verbs expressing attachment) and *wan* ‘finish’. RVCs are not regarded as aspectual heads. The higher layer is further split into two: Outer Aspect holds elements that can associate with T, and Middle Aspect being holds aspectual suffixes that exhibit incompleteness effects due to failure to associate with T. The dividing lines of the three aspects are on *vP* and *VP*.

- (126) $[_{TP} \dots [_{AspP1} zai, -guo [_{VP} \dots [_{AspP2} -zhe^1, -le \dots [_{VP} V-Asp3=PCs$

B.2 Inner aspect in Cantonese

The list of inner aspectual elements in Cantonese is given below. (a-h) denote the endpoint of events with varying meanings such as event achievement, completion, successful perception, etc. (i-j) also make reference to endpoints, but they encode an additional salient ‘in advance’ meaning. (k) mainly expresses an adversative meaning with a requirement on achieved events. (l-m) concern the internal intervals of events and convey continuation (see Yip 2025 for extensive discussion, and Sio:2025 for a follow-up study).

(127) Postverbal inner aspectual elements in Cantonese

(Cheung 1972, 2007:113-115, Matthews & Yip 1994:210-221, S.-W. Tang 2015:71-89)

- a. *zoek6* ‘be on target’, e.g. *fan3-zoek6* ‘slept’
- b. *dou2* ‘arrive’, e.g. *wan2-dou2* ‘found’ (cf. J. Huang 2021)
- c. *gin3* ‘see (vs. look)’, e.g. *teng1-gin3* ‘hear (vs. listen)’
- d. *hou2* ‘good’, e.g. *zou6-hou2* ‘being done’
- e. *dim6* ‘all right’, e.g. *gaau2-dim6* ‘finished, settled’
- f. *jyun4* ‘finish’, e.g. *sik6-jyun4* ‘finished eating’
- g. *seng4* ‘succeed’, e.g. *joek3-seng4* ‘succeeded in making an appointment’
- h. *hei2* ‘finish, ready, (lit.) lift’, e.g. *waak6-hei2* ‘finished drawing’
- i. *lok6* ‘finish (a long time ago), (lit.) fall’, e.g. *zyu2-lok6* ‘taught way back then’
- j. *ding6* ‘in advance’, e.g. *zyu2-ding6* ‘cooked in advance’ (cf. Wong 2018)
- k. *can1* ‘adversative’, e.g. *da-can* ‘injure by hitting’
 (with achievement requirements, cf. Sio 2020)
- l. *zyu6* ‘hold’, e.g. *kam2-zyu6* ‘cover still’
- m. *sat6* ‘firm’, e.g. *mong6-sat6* ‘keep looking’

While most of them are “phase complements”, some may not be labeled so in the original sources. S.-W. Tang 2015, for example, treats *lok6*, *ding6*, *can1*, *zyu6* and *sat6* as verbal suffixes.

We employ three tests for inner aspects: (i) suffixation of outer aspectual affixes like perfective *-zo*; (ii) *-dak/m-* ‘able/not’ infixation; (iii) following an RVC.

We take the first test as the crucial diagnostics for distinguishing inner aspect from outer aspectual suffixes. All the inner aspect markers can be suffixed by the perfective *-zo*. Note that this test does not provide evidence for the morphological status (i.e. whether these postverbal elements are verbal complement or suffixes), since suffix stacking is possible in Cantonese (S.-W. Tang 2015). Instead, this test shows that inner aspectual elements are lower than *-zo* in the structure (and thus closer to the root by Mirror Principle, Baker 1985).

- (128) *-zo* ‘PFV’ suffixation (J. Huang 2021:4)
- Aaming lou-dou-zo jat-go offer.
 Ming get-PC-PFV one-CLF offer
 ‘Ming got an offer.’

The second test is *-dak/m-* ‘able/not’ infixation, which is a typical property displayed by RVCs.⁵¹ The test provides evidence for the morphological status of these elements being PCs as opposed to aspectual suffixes.

- (129) *-dak/m-* ‘able/not’ infixation (Video caption, 2021-2-10)
- jausi dou lo-m-dou, houi gamjat gam-do jan, zau meibit lo-dak-dou
 sometimes also get-not-PC like today that-many person then uncertain get-able-PC
 ‘Sometimes (you) cannot get it. When there are as many people like today, it’s not certain that (you) can get it.’

The third test is whether the postverbal elements may follow a RVC, which has been discussed above. This test effectively distinguishes these inner aspectual elements from RVCs since RVC stacking is not possible in Cantonese.

The results are summarized below. We treat the first two rows as PCs, and the last two rows as verbal suffixes.⁵²

Table 6 Grammatical properties of inner aspectual elements in Cantonese

Inner aspectual elements	<i>-zo</i> suffixing	<i>-dak/m-</i> infixing	following RVCs
<i>-dou2</i> , <i>-jyun4</i> , <i>-zyu6</i>	YES	YES	YES
<i>-zoek3</i> , <i>-gin3</i> , <i>-hou2</i> , <i>-dim2</i> , <i>-seng4</i> , <i>-can1</i> , <i>-sat6</i>	YES	YES	NO
<i>-ding6</i>	YES	NO	YES
<i>-lok6</i> , <i>-hei2</i>	YES	NO	NO

⁵¹ Except universal *-saai*, which is arguably a suffix but it can also take *-dak/m-* ‘able/not’ infixation (P. P.-l. Lee 2012).

⁵² That *-lok6* and *-hei2* cannot follow RVCs might be due to their preference for a monosyllabic host verb.

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